

DESIGNING ZERO-PENALTY SETS: FROM CONDITIONAL VARIATIONAL BOUNDS TO FIXED PROTOTYPE REGULARIZATION

Anonymous authors

Paper under double-blind review

ABSTRACT

Conditional information bottlenecks identify the *information point* of a representation — CEB’s minimum necessary information (MNI) — but not the *geometry* at that point: we prove that any MNI solution can be bijectively recoded into an equivalent solution whose inter-class distances, scale, and conditioning are arbitrary, so the optimizer, not the objective, settles the final label geometry; the variational certificate moreover constrains the noisy training code, not the deterministic representation inference deploys. We take the zero-penalty set of the regularizer as the design object and propose the Fixed-Geometry Information Bottleneck (FGIB): the inference path stays deterministic, and the prior is replaced by a *frozen* label-conditional geometry — an orthonormal anchor frame for classification, fixed random-Fourier anchors for regression — reached through a side head inference never reads. Because the reference cannot move, we prove the variational KL is *identically* a fixed-prototype center loss throughout training: a deterministic main path freezes the zero-initialized variance head at the prior value, so the compression term is exactly $\frac{1}{2\tau^2} \|\mu - av_y\|^2$ at every step — the variational form carries the conditional-information interpretation, while the empirical mechanism is pure geometry regularization. In the canonical configuration the side head is a *fixed orthogonal* map, anchoring the deployed representation directly with $\kappa(A^\top A) = 1$, and at a fixed classifier scale every β selects a distinct interior optimum whose fully compressed corner is the class-separated anchor configuration (cross-entropy $\ln(1 + (K-1)e^{-ca})$, not the chance-level $\ln K$ of a marginal prior). Across seven classification settings and four backbones under equal budget, FGIB’s β knob is the only one that is non-destructive over a grid spanning five orders of magnitude — a property that survives a scale-free comparison of effective regularization strength — while its accuracy is essentially the backbone’s: gains are small and statistically unstable (classification-only Wilcoxon $p \approx 0.22$). The contribution is the designed, well-conditioned, predictive geometry; a geometry ablation (orthonormal vs. simplex/ETF vs. random frames vs. learned centers, on the same representation) shows fixedness matters more than the frame, and the fixed head’s measured price is quantified rather than hidden. Regression is an exploratory transfer kept in the appendix; a full ledger separates what is proved from what is open.

1 INTRODUCTION

An information bottleneck asks a representation Z of an input X to be maximally informative about a target Y and minimally informative about everything else (Tishby et al., 1999). The deep variational instantiation (Alemi et al., 2017) replaces the intractable $I(X; Z)$ with $\mathbb{E} \text{KL}(q(z|x) \| m(z))$ and minimizes $\text{CE} + \beta \text{KL}$; the Conditional Entropy Bottleneck (Fischer, 2020) sharpens the target to the *residual* $I(X; Z|Y)$ with a learned backward encoder $b(z|y)$. Two properties of that construction are usually left implicit. First, *where the zero-penalty set lives*: the KL vanishes exactly on the prior, so the prior chooses the set of representations the regularizer never penalizes — a marginal prior places it at collapse, a learned conditional prior leaves it an underdetermined family, and the optimizer, not the objective, selects the final label geometry. Second, *the object being regularized*

is not the object being deployed: the bound constrains a noisy training code $z \sim q(z|x)$ while inference consumes the deterministic mean $\mu(x)$. This paper takes the zero-penalty set as the design object, and asks what changes if it is fixed by design rather than left to move or to vanish.

Conditional bottlenecks identify information, not geometry. In the deterministic scenario CEB’s objective is, up to constants, the IB Lagrangian at $\beta_{\text{IB}} = 1 + \gamma > 1$, and every $\gamma > 0$ shares the same exact minimizer — the copy corner — which is precisely CEB’s own declared target, the MNI point (Proposition 1, Remark 1). Read correctly, this is not a failure of CEB: the objective does what it is declared to do — identify the information point. What it does *not* identify is the geometry of the representation at that point.

Geometric non-identifiability at the MNI solution. The gap is made precise by a single proposition, the paper’s theoretical pivot. (G) *Geometric non-identifiability*: any MNI solution can be bijectively recoded into an equivalent solution with the same information point and the same optimal Bayes cross-entropy, while its Euclidean geometry — inter-class distances, scale, condition number — is arbitrary within the class: for linear recodings, $\|B(m_y - m_{y'})\|^2 = (m_y - m_{y'})^\top B^\top B (m_y - m_{y'})$ with $\kappa(B^\top B)$ arbitrary (Proposition 2). The objective identifies an *equivalence class*, never a member of it; which member — which geometry — is reached is settled by the optimizer. Two further observations are supporting rather than load-bearing: the residual certificate is blind to label retention on the collapse-to-copy manifold (Proposition 3), and no finite certificate bounds the residual of a deployed deterministic representation (Proposition 5) — a limitation of every variational bottleneck, ours included. A variance race of the Gaussian parametrization, which exists only when the prior variance is trainable (CEB, not VIB; Remark 5), is measured as an optimization diagnostic in the appendix and carries no load in the argument.

Designing the zero-penalty set. FGIB (Section 3) makes three changes that follow from the diagnosis: (i) the inference path is deterministic — the classifier consumes the encoder output h at training *and* at evaluation; (ii) the stochastic bottleneck moves to a *side path* ($\mu(x) = Ah(x)$ plus a log-variance head) that no inference path ever reads, keeping the variational bound finite without putting noise in front of the classifier; (iii) the label-conditional prior is *fixed by design*, $r(z|y) = \mathcal{N}(av_y, \tau^2 I)$ with an orthonormal anchor frame, so the zero-penalty set is the prescribed class-separated configuration rather than collapse or an underdetermined family. The *canonical* configuration fixes the side head as an orthogonal map ($A^\top A = I$), anchoring the deployed representation directly with $\kappa = 1$ by construction (Proposition 7); a trainable side head is an optional empirical variant whose conditioning is measured, not assumed (Section 4.5).

Freezing the reference is what buys the theory. Against a fixed μ_p , the KL’s displacement term $\frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ is *quadratic* in the alignment scalar — the convexification squared-IB (Kolchinsky et al., 2019a) installs by hand is built in — and at fixed classifier scale every β has a unique interior optimum on a strictly concave trade-off curve (Propositions 9 and 10). Equally important, the KL now vanishes at the class-separated configuration $\mu_y = av_y$, with cross-entropy $\ln(1 + (K-1)e^{-ca}) \rightarrow 0$ as a grows, where a marginal $\mathcal{N}(0, I)$ prior places KL = 0 at $\mu \equiv 0$ with cross-entropy $\ln K$ — chance (Remark 15): the degenerate corner is excluded by geometry rather than by the optimizer’s good behavior.

Fixed geometry as a variationally interpretable prototype regularizer. We evaluate the canonical configuration on 9 datasets over MLP, CNN, GCN and LSTM backbones against the deterministic backbone and five objectives: VIB (Alemi et al., 2017), squared-IB (Kolchinsky et al., 2019a), nonlinear IB (Kolchinsky et al., 2019b), CEB (Fischer, 2020) and supervised β -DVCCA (Abdelaleem et al., 2025); the sweep is 1,020 configurations $\times$ 5 seeds = 5,100 runs plus equal-budget and ablation controls, with every hyperparameter selected on validation. Three findings, in Section 4:

- *The regularizer is exactly a fixed-prototype loss, and the prototype geometry is designed.* A deterministic main path freezes the zero-initialized variance head at the prior value, so the variational KL is *identically* $\frac{1}{2\tau^2} \|\mu - av_y\|^2$ at every training step (Proposition 8): the variational form carries the conditional-information interpretation, the empirical mechanism is pure geometry regularization, and a geometry ablation shows the stability rides on *fixedness and class separation*, not on the particular frame (Table 10).

- *The canonical fixed head costs little and closes the theory.* Across the seven classification settings under equal budget, FGIB-H (at the re-tuned scale $a=4$) stays within 0.23 accuracy points of its backbone at its worst setting and is best or tied-best on 2 of 7 settings, with mean rank 2.86 of 5 against the backbone’s 3.29 and $\kappa(A^\top A) = 1$ by construction (Table 9); the optional trainable head reaches the same accuracies but its conditioning degrades to $10^{5.8}$ – $10^{6.8}$ without buying accuracy.
- *The knob is non-destructive, and this survives a scale-free comparison.* Sweeping β over a six-point grid spanning five orders of magnitude leaves FGIB within 0.52 accuracy points of its backbone on *every* classification setting, while VIB, CEB and DVCCA fall to the trivial predictor on all seven (Table 3); the same picture holds on the scale-free axis of effective regularization strength (Table 11, Figure 2). Accuracy gains are small and task-set-sensitive (classification-only Wilcoxon $p \approx 0.22$), so we claim a designed, well-conditioned, predictive geometry — not a better stochastic bottleneck.

What we do not claim. The one-to-one sweep requires a bounded classifier scale the implementation does not enforce (Remark 14); no finite certificate bounds the deployed deterministic residual, ours included (Proposition 5); the sweep is a KL–CE surrogate curve on a class-symmetric family, not a recovered IB curve (Remark 13); and the KL adds the residual-information reading and the sweep theory, not a measured accuracy margin. Appendix A keeps a complete ledger of what is proved and what is open.

2 PRELIMINARIES: CONDITIONAL BOTTLENECKS IDENTIFY INFORMATION, NOT GEOMETRY

X is the input, Y the label with K classes, and an encoder induces the Markov chain $Y - X - Z$; we use the loss convention $\text{CE} + \beta \text{KL}$ ($\beta = 1/\gamma$, $\beta_{\text{IB}} = 1 + 1/\beta$). The variational objects $R = \mathbb{E}[\text{KL}(q(z|x)||m(z))]\geq I(X;Z)$ (VIB’s rate) and $\text{ReX} = \mathbb{E}[\text{KL}(q(z|x)||b_\theta(z|y))]\geq I(X;Z|Y)$ (CEB’s residual, trainable b_θ) — and ReX_F with a *fixed* anchor r — are instances of the gap identity $\mathbb{E}[\text{KL}(q(z|x)||r(z|y))]=I(X;Z|Y)+\mathbb{E}_y[\text{KL}(q(z|y)||r(z|y))]$ (Lemma 4): a bound plus a slack, and everything below turns on *who is allowed to move that slack to zero*. Two observations follow (proved in Appendix A, stated in detail in Appendix B).

The information point is unique. In the deterministic scenario every $\gamma > 0$ of CEB has the same exact minimizer, the copy corner — which is the MNI point (Proposition 1, Remark 1). This is not a failure of CEB: the objective does what it is declared to do, identify the information point. What it does not identify is the geometry at that point, and γ cannot be reading off a position on the IB curve.

Result 1 (Geometric non-identifiability at the MNI solution). *If Z attains the MNI point, then every bijective recoding $\psi(Z)$ attains the same information point, $I(X;\psi(Z)|Y)=I(X;Z|Y)$ and $I(Y;\psi(Z))=I(Y;Z)$, with the same optimal Bayes cross-entropy $H(Y|\psi(Z))=H(Y|Z)$. The Euclidean geometry is unconstrained within the class: for linear recodings $B \in \text{GL}(d)$ the class-mean distances transform as $\|B(m_y - m_{y'})\|^2 = (m_y - m_{y'})^\top B^\top B (m_y - m_{y'})$, so $B = sI$ rescales every inter-class distance by s and $B = \text{diag}(c, 1, \dots, 1)$ sends $\kappa(B^\top B) \rightarrow \infty$. Inter-class distances, scale and conditioning of an MNI solution are optimizer artifacts, not objective-selected quantities (Proposition 2).*

The certificate is blind on the necessity axis. ReX is identically 0 on the collapse-to-copy manifold T_α from total collapse to full copy, and the objective trades one nat of label information for $1/\beta$ nats of certified compression (Proposition 3). Since the full objective also contains the prediction term, this is a rate statement about the certificate — supporting, not load-bearing.

Scope. The certified object is the noisy training code, never the deployed representation: $I(X;Z|Y) \leq I(X;\mu(X)|Y) = +\infty$ for any continuous non-class-constant μ , CEB’s or FGIB’s (Proposition 5). A variance race of the Gaussian parametrization exists only when the prior variance is trainable (CEB, not VIB, whose fixed prior restores the posterior variance and whose high- β failure is the mean collapse $q \rightarrow \mathcal{N}(0, I)$; Remark 5); it is measured as a diagnostic in the appendix and carries no load in the argument.

3 METHOD: THE FIXED-GEOMETRY INFORMATION BOTTLENECK

3.1 MODEL

A shared encoder computes the deterministic representation $h = f_{\text{enc}}(X)$, and the task head consumes h directly at training and evaluation: $\text{logits} = c(h)$. In parallel, a *side path* maps h to a diagonal Gaussian posterior

$$q(z|x) = \mathcal{N}(\mu(x), \Sigma(x)), \quad \mu(x) = Ah(x), \quad \Sigma(x) = \text{diag} \exp(\ell(h(x))), \quad (1)$$

with a linear mean head A and a log-variance head ℓ , neither evaluated at inference. The label-conditional prior is a set of frozen buffers,

$$r(z|y) = \mathcal{N}(av_y, \tau^2 I), \quad V = [v_1, \dots, v_K] \text{ orthonormal}, \quad V^\top V = I_K, \quad (2)$$

where V is the orthonormal factor of the QR decomposition of a random matrix, $a > 0$ the *anchor scale* and τ^2 the *anchor variance* (default 1). The training objective is

$$L_{\text{FGIB}} = \mathbb{E}[\text{CE}(c(h), y)] + \beta \mathbb{E}[\text{KL}(q(z|x) \| r(z|y))]. \quad (3)$$

Gradients from the second term reach the main path only through the shared encoder. The log-variance head is zero-initialized, and Result 3 proves the variance stays pinned at τ^2 for the entire run, so the realized regularizer is exactly $\frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ at every step. The canonical configuration fixes the side head to an *orthogonal* map — with $\dim z = \dim h$ one may take $A = I$ without loss (Result 2) — so the anchor matching acts directly on the deployed representation; the trainable head of Eq. (1) is then an optional ablation, and we call the canonical form *FGIB-H* ($A = I$). For regression the per-class table is replaced by fixed random-Fourier anchors (Rahimi & Recht, 2008) on a sphere of radius a ; $a = 0$ recovers a shared $\mathcal{N}(0, \tau^2 I)$ prior (Remark 24). Three design decisions distinguish this from CEB: the classifier consumes h rather than z ; the two loss terms act on disjoint parameter sets; and the reference is a frozen buffer rather than a network.

Theory–implementation layering. The analysis idealizes the side head as a linear invertible map; the implementation uses an `nn.Linear` with *bias* and no conditioning constraint — and that freedom is what lets the trainable head absorb the anchor fit without moving h (Section 3.3(a), measured in Section 4.5). The canonical FGIB-H removes the gap rather than flagging it: $A = I$ is orthogonal by construction, so the idealizations — invertibility, conditioning, the pull-back reference of Result 2 — hold exactly in the implementation. A second gap: the exact KL invariance of Result 2 holds for the underlying Gaussians, but the diagonal family is not closed under A^{-1} , so within it the identity is exact distributionally and approximate as a variational identity (Remark 10).

3.2 WHAT THE FROZEN REFERENCE BUYS

Result 2 (The side path is a CEB bottleneck on the deployed representation). *Let $\dim z = \dim h$ and A be invertible. Pulling back by A^{-1} maps the posterior to $q'(\tilde{z}|x) = \mathcal{N}(h(x), A^{-1}\Sigma A^{-\top})$ — centered at the representation inference uses — and the anchors to $\mathcal{N}(aA^{-1}v_y, \tau^2(A^\top A)^{-1})$. The Gaussian KL is invariant under a common invertible linear map (Lemma 1), so the regularizer is pointwise identical to a CEB bottleneck — backward encoder frozen at the current side head — applied to $\tilde{z} = h + \tilde{\varepsilon}$ (Proposition 6); at the variance optimum the perturbation is $\tau(A^\top A)^{-1/2}\varepsilon$: known, side-path-only noise. This licenses calling the construction a bottleneck at all: the regularizer carries a residual-information bound centered on the deployed representation — an interpretation a bare prototype loss lacks. The identity holds at each fixed A ; because a trainable A moves, the pulled-back reference moves with it. The canonical configuration removes this channel: for a fixed orthogonal A , $\|Ah - av_y\| = \|h - aA^\top v_y\|$, so the regularizer is a fixed-anchor center loss on the deployed representation itself, with $A^\top A = I$ and $\kappa = 1$ at every step; for Haar-distributed random orthogonal A the rotated frame is identically distributed to $\{v_y\}$ across seeds, so $A = I$ is a representative canonical choice (Propositions 7, 13). We call this form FGIB-H and measure its price across settings (Section 4.5, Table 9).*

Result 3 (A deterministic main path freezes the variance: the realized KL is a fixed-anchor center loss). *With the classifier consuming h (never z), a zero-initialized log-variance head, a frozen prior log-variance $\ell_p = \ln \tau^2 = 0$, and a first-order optimizer without weight decay, the posterior log-variance is identically $\ell_q \equiv \ell_p$ at every training step, and the regularizer is exactly $\text{KL}(q \| r) =$*

$\frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ — the KL of Result 4 evaluated permanently at the variance optimum, not merely in the limit. The mechanism is exact invariance, not attraction: the per-coordinate partial derivative $\partial_{\ell_q} \text{KL} = \frac{1}{2}(e^{\ell_q - \ell_p} - 1)$ vanishes at $\ell_q = \ell_p$, so the zero-initialized head receives zero gradient at every step — we make no global-attractor claim for arbitrary initializations (Proposition 8). For trainable-prior bottlenecks the released variance is a real channel — CEB’s parametrization races toward the clamp (Remark 5) — and the deterministic main path eliminates it by design. The same argument pins the posterior variance in every FGIB-family ablation: FGIB, TAFGIB and CentFGIB are exactly the same loss up to the trainability of the anchor means. The one variant that escapes the family, DCEB’s trainable prior variance, is the only one that collapses at $\beta = 10$; its measured trajectory confirms the corrected sign of Corollary 1 (the prior log-variance rises at $\beta=1$, never falls), while the collapse itself occurs with the variance near initialization — an open, measured diagnostic (Section 4.5).

Result 4 (Every β selects a distinct interior optimum). Consider the class-symmetric aligned family — the encoder maps class y to $h_y = s v_y$, the classifier is held at $w_j = c v_j$ with fixed scale c . Then

$$\text{KL}(s, \sigma^2) = \frac{\dim z}{2} \left(\frac{\sigma^2}{\tau^2} - 1 - \ln \frac{\sigma^2}{\tau^2} \right) + \frac{1}{2\tau^2} (s - a)^2, \quad \text{CE}(s) = \ln(1 + (K - 1)e^{-cs}), \quad (4)$$

and the variance optimum is $\sigma^{2*} = \tau^2$ for every β , because the prediction term does not see σ^2 (Lemma 3). The Lagrangian $L_\beta(s) = \text{CE}(s) + \frac{\beta}{2\tau^2}(s - a)^2$ is then strictly convex with a unique minimizer $s^*(\beta) > a$, strictly decreasing in β with $s^* \rightarrow a$ as $\beta \rightarrow \infty$, so $\beta \mapsto (\text{KL}(s^*), \text{CE}(s^*))$ is one-to-one onto the entire curve, which is strictly concave (Propositions 9, 10). The mechanism is the point of freezing the reference: deterministic-IB has no usable compression knob, repaired in Kolchinsky et al. (2019a) by squaring the compression term by hand; FGIB needs no surgery, because $\text{KL} = \frac{1}{2\tau^2} \|\mu - \mu_p\|^2 + (\text{variance terms})$ is quadratic in the displacement whenever μ_p does not move: a fixed anchor turns self-referential entropy compression into a squared distance, and the convexification is inherited, not installed.

Result 5 (A persistent force and a non-collapsed compressed endpoint). At a fixed encoder, CEB’s bound is tighter in value: $\text{ReX}(\theta^*) \leq \text{ReX}_F$, the excess being the anchor gap $\mathbb{E}_y[\text{KL}(q(z|y) \|\mathcal{N}(av_y, \tau^2 I))] - \mathbb{E}_y[\text{KL}(q(z|y) \|\mathcal{N}(m_y, S_y))] \geq 0$ (Proposition 11). That excess is the price of the fixed anchor, paid for a reason: CEB’s gap can be driven to zero with the representation untouched, whereas FGIB’s can shrink only if the posterior moves toward the anchors — the force is persistent (Remark 19). Moreover, the $\text{KL} = 0$ point is $\mu_y = av_y$, a class-separated configuration with $\text{CE} = \ln(1 + (K-1)e^{-ca})$, bounded and $\rightarrow 0$ as a grows; a marginal $\mathcal{N}(0, I)$ prior places $\text{KL} = 0$ at $\mu \equiv 0$ with $\text{CE} = \ln K$, i.e. chance (Remark 15). This is the prediction the experiments test most directly: pushing β up should not destroy an FGIB model, and a is the dial controlling how benign the compressed corner is (Tables 3 and 2).

3.3 WHAT THE FROZEN REFERENCE DOES NOT BUY

We state the open items here because two of them bear on how the experiments should be read. (a) A trainable side head is a second trainable party in the KL and can absorb the anchor fit without moving h ; the canonical fixed head closes this by construction, and the measurement shows the trainable variant’s $\kappa(A^\top A)$ degrades to $10^{5.8}$ – $10^{6.8}$ while accuracy is unaffected — the freedom buys nothing. (b) The sweep needs a bounded classifier scale, which the implementation leaves free, so Result 4 is conditional as implemented. (c) The certificate still does not bound the deployed residual. (d) The compressed endpoint is designed, not certified: the anchor corner is a label code, not semantic compression (Remark 15). (e) The sweep is over a surrogate, the $(\text{KL}_{\text{anchor}}, \text{CE})$ plane, not the information plane (Remark 13). Full statements in Appendix C.

4 EXPERIMENTS

4.1 SETUP

Tasks and backbones. We use 9 datasets over 4 backbones, giving 12 task×backbone settings: MNIST (LeCun et al., 1998) (MLP/CNN), ImageNet-100 (Deng et al., 2009) frozen ResNet-50 (He et al., 2016) features (MLP/CNN), Cora (Sen et al., 2008) (GCN (Kipf & Welling, 2017)), IMDb (Maas et al., 2011) and AG News (Zhang et al., 2015) (frozen BERT (Devlin et al., 2019) token features) with LSTM (Hochreiter & Schmidhuber, 1997) encoders; and the regressions California

Table 1: Test accuracy (%) of the validation-selected configuration of each objective on the seven classification settings, averaged over 5 runs (seeds 0–4); higher is better. *Det.* is the deterministic backbone trained with the task loss alone. For every bottleneck β is chosen per cell by mean best-epoch validation score — and for FGIB the anchor scale a as well — never by test score, so FGIB is selected over a 6×9 grid and the others over a 6-point grid; exact ties are broken by a second validation quantity and then by fixed grid order. The FGIB column is the optional trainable-head variant; the canonical fixed head (FGIB-H) is Table 9. The five regression settings are Table 5, reported as exploratory. Best per row in bold (ties within 0.01 points both bolded); per-run standard deviations and the selected hyperparameters are in Table 4, whose last column reports the equal-budget FGIB control.

Task	Backbone	Det.	VIB	SVIB	NIB	CEB	DVCCA	FGIB
MNIST	MLP	98.33	98.28	98.10	98.26	98.28	98.24	98.49
MNIST	CNN	99.13	99.28	99.19	99.19	99.25	99.28	99.27
ImageNet-100	MLP	83.12	82.65	83.02	82.34	82.56	82.56	83.96
ImageNet-100	CNN	83.40	82.94	82.61	82.72	82.33	83.09	83.55
Cora	GCN	87.68	86.83	77.56	86.35	86.27	86.83	87.68
IMDb	LSTM	84.85	85.43	85.99	86.11	85.55	85.69	86.04
AG News	LSTM	92.94	92.61	92.70	92.91	92.66	92.78	92.71
Mean rank (of 7)		3.14	4.00	5.14	4.29	5.14	3.57	1.86
Best or tied-best (of 7)		2	1	0	1	0	1	4

Housing (Pace & Barry, 1997) (MLP), STS-B (Cer et al., 2017; Wang et al., 2019) (GloVe (Pennington et al., 2014) LSTM), ZINC-12k (Dwivedi et al., 2023) (GCN), and AgeDB (Moschoglou et al., 2017) (MLP/CNN). Splits are 60/20/20 (seed 42) except ZINC, so every comparison is internal to the same split; a subject-disjoint AgeDB control preserves the objective ordering (Table 12). Appendix G gives the preprocessing.

Objectives. Each backbone carries the same seven heads: the deterministic backbone, VIB (Alemi et al., 2017), squared-IB (Kolchinsky et al., 2019a, SVIB, penalty βKL^2), nonlinear IB (Kolchinsky et al., 2019b, NIB, non-parametric pairwise bound, learned noise variance), CEB (Fischer, 2020, diagonal-Gaussian backward encoder), supervised β -DVCCA (Abdelaleem et al., 2025, VIB plus an input-reconstruction decoder), and FGIB. For regression, CEB and FGIB condition their priors on the min-max-scaled continuous target. Every variant shares the encoder, the bottleneck dimension ($\dim z = 256$) and the optimizer. The objectives’ compression terms are not dimensionally identical (NIB’s bound is in nats; DVCCA’s β also weights reconstruction), so β is a per-objective knob swept independently and selected on validation, never a shared axis of equal compression.

Protocol. Adam (Kingma & Ba, 2015) at 10^{-3} (10^{-2} for Cora), batch size 512 (128 for ZINC), at most 100 epochs, early stopping with patience 10 on validation macro-AUC (classification) or validation R^2 (regression); each configuration is run 5 times with seeds 0..4 and we report the mean over runs. The grid is $\beta \in \{10^{-4}, \dots, 10\}$ for every bottleneck plus $a \in \{1, \dots, 16\}$ for FGIB: 1,020 configurations, 5,100 runs. **Hyperparameters are selected on validation, never on test:** each cell reports the test score of its best mean-validation configuration, with exact ties broken by a second validation quantity and then fixed grid order, so no selection step consults the test set. Per-run standard deviations and selected hyperparameters are in Appendix D; the ablations of Section 4.5 use the identical protocol on MNIST/MLP.

4.2 ACCURACY: FGIB RANKS FIRST, BUT NO BOTTLENECK WINS BY MUCH

Table 1 reports the validation-selected configuration of each objective on the seven classification settings (the five regression settings are Table 5, reported as exploratory). Its FGIB column is the *optional trainable-head* variant, selected over the full 6×9 grid; the canonical fixed-head configuration (FGIB-H), selected over the same 6-point budget as every other objective, is Table 9, whose rank and win counts are computed over the complete seven classification rows. With that said: the trainable variant attains the best mean rank, 1.86 of 7, and is best or tied-best on 4 of 7 settings;

the runner-up is the deterministic backbone at 3.14 and every prior bottleneck sits above 3.5; on the regression rows its rank is 1.60 (Table 5).

Three qualifications are essential. First, *the margins are small*: excluding the two settings where every bottleneck helps a weak backbone, no objective improves on its backbone by more than about one point, and losses can be large even at the validation-selected optimum (SVIB gives up 10.1 points on Cora). Reading Table 1 as evidence that bottlenecks substantially improve in-distribution accuracy is reading it wrong; it shows that FGIB is the only one that is reliably not *worse*. Second, *FGIB’s selection grid is 9× larger*: the equal-budget control (a single fixed $a = 16$, post-hoc) has mean rank 2.14 on classification (2.33 over all 12), still ahead of the backbone and every prior bottleneck — part of the rank advantage is search budget; the ordering of methods is not (Table 7: the fixed-anchor rank ranges from 2.00 at $a = 14$ to 3.17 at $a = 1$). Third, *paired statistics, with their sensitivity reported* (Table 6): a Wilcoxon test over the 12 setting-level paired means finds FGIB the only objective whose gain over its backbone is significantly positive ($p = 0.014$; $p = 0.12$ at the equal-budget control), and FGIB beats each prior bottleneck pairwise ($p \leq 0.016$) — but neither subset is significant on its own (classification-only $p \approx 0.22$, mean gain +0.32 points; regression-only $p \approx 0.13$, mean +3.05), and excluding the two AgeDB rows the mixed p falls to 0.055 with the mean gain falling from +1.46 to +0.62 points. The mixed $p = 0.014$ pools heterogeneous metrics across 12 non-independent settings and is exploratory only. The data therefore support *non-destructiveness on classification* and a small, unstable gain overall — not a significant in-distribution accuracy improvement; we take the *rank and the non-destructiveness of the knob* as the primary statistical claims.

4.3 THE KNOB: A FIXED-PROTOCOL STRESS TEST, RE-READ ON A SCALE-FREE AXIS

Result 5 predicts what happens at the *other* end of the sweep: a marginal-prior bottleneck has its $KL = 0$ point at $\mu \equiv 0$, so raising β far enough must drive it to the trivial predictor; FGIB’s $KL = 0$ point is class-separated. Table 3 reports the stress test: for each objective we take the *worst* test score produced by any β in the grid, and report the deficit against the deterministic backbone of the same row — a shared *protocol*, deliberately not a shared axis of effective strength, since the compression terms are not dimensionally identical (Section 4). The separation is stark. VIB, CEB and DVCCA are driven below the trivial predictor on all 7 classification settings (median worst-case deficits 71.9, 66.1, 79.1 points); SVIB is only marginally better (64.8): squaring the KL does not move the corner (Remark 15). FGIB’s median worst-case deficit is 0.31 points at the validation-selected anchor and 0.35 at fixed $a = 16$; on classification the worst case over the six-point, five-order-of-magnitude β grid is within 0.52 points of the backbone on *every* one of the seven settings (per-setting values in Table 3), against deficits of 34–88 points for VIB, CEB and DVCCA.

A scale-free axis of effective strength. Because the penalties are not dimensionally identical, we re-measure the stress test on the axis of *effective regularization strength* $r = \beta \|\nabla_{\theta} R\|_2 / \|\nabla_{\theta} CE\|_2$ at a shared one-epoch CE-only operating point on MNIST/MLP (Table 11; Figure 2). The axis exposes two confounds and one fact that survives. (i) *NIB’s stability is inertia*: its bound’s gradient is numerically zero across the entire grid ($r \equiv 0$), the saturating-gradient phenomenon of Kolchinsky et al. (2019b, §IV A) made quantitative. (ii) *SVIB’s raw-grid collapse is a scale artifact*: its penalty βKL^2 carries different units, and at raw $\beta = 1$ its effective strength is $r \approx 10^{6.2}$ — two orders of magnitude beyond any KL-nat objective’s grid — so its 38-point deficit there does not signal fragility relative to the others; squaring the KL rescales the grid, not the geometry. (iii) *The fact that survives*: among the KL-nat objectives, the others collapse between $r \approx 10^{3.15}$ (CEB) and $10^{3.23}$ (VIB, DVCCA), while FGIB($a=16$) reaches $r \approx 10^{3.75}$ — $\approx 3\times$ beyond the largest collapse threshold — with zero deficit across the whole range: on the effective-strength axis the raw grid pushes FGIB *further* than any objective that collapses, and it does not break.

NIB is the interesting control. It is as stable as FGIB (median deficit 0.53), but for a different reason: its non-parametric bound saturates with exponentially vanishing gradient (Kolchinsky et al., 2019b, §IV A), so the knob barely acts — on MNIST/MLP its accuracy moves from 98.15 to 98.27 across the entire sweep. Two gain summaries in Table 3 make the distinction: at the *best* β (test-selected, an exploratory upper bound) NIB scores −0.27 points, and VIB, SVIB, CEB, DVCCA are likewise negative — only FGIB is positive (+0.39; +0.27 at $a = 16$); at the *validation-selected* configuration

FGIB scores +0.26 against -0.21 to -0.37 for the others. *Among the six bottlenecks, FGIB is the only one whose compression knob is both non-destructive and, at the median, worth turning.*

The exceptions are informative and we do not hide them: FGIB is destroyed (worst case > 5 points below the backbone) in 3 of 12 settings at $a = 16$, and all three are regression — STS-B (13.2), AgeDB (15.6 CNN, 24.0 MLP), with ZINC just below threshold at 4.5. We therefore report the regression results as *exploratory* and keep the core claims on classification; the theory anticipates this asymmetry — the random-Fourier anchor construction for continuous targets is a transfer argument, not a proof (Remark 18).

4.4 THE ANCHOR SCALE IS THE DIAL THE THEORY SAYS IT IS

Result 5 says when robustness should fail: the compressed corner costs $CE = \ln(1 + (K-1)e^{-ca})$, small only when ca is large relative to $\ln(K-1)$; at $a \rightarrow 0$ the anchors merge onto a single origin and FGIB degenerates to a VIB-style marginal prior (Remark 24). Table 2 holds β at the largest grid value and varies a . The prediction holds: at $a = 1$ FGIB behaves like the marginal-prior bottlenecks (1.0% on ImageNet-100/MLP, 26.6% on Cora), and growing a recovers the backbone — monotonically on ImageNet-100/MLP, with local dips elsewhere. The trend, not the monotonicity, is what the theory predicts, and it holds on 10 of 12 rows; the two exceptions are the two the theory says should have none: IMDb is binary (the corner is already cheap at $a = 1$), and ZINC is regression, where the random-Fourier anchors carry no corresponding guarantee (Remark 18). Consistent with the $(K-1)$ factor, the 100-class task needs by far the largest anchor.

The formula does not predict the transition threshold, and we do not pretend it does: Cora ($K=7$) collapses at $a \leq 2$ where MNIST ($K=10$) does not, since the classifier scale c enters the corner value multiplicatively and is free in our implementation (Remark 14); Table 2 establishes the qualitative mechanism, not a quantitative fit.

4.5 CONFOUND ABLATIONS AND DIAGNOSTICS

This section separates FGIB’s two design changes (deterministic main path; frozen prior) and verifies the closure results. All runs use the main protocol, except the 400-epoch controls; the additional experiments total 710 runs (full per-variant details in Appendix E).

What the ablations show. VIB and CEB fall to chance at $\beta = 1$ (15.3% and 26.8%); *DCEB* — the deterministic path with CEB’s trainable backward encoder as prior — holds at ≈ 98 through $\beta = 1$ and falls to 11.4% only at $\beta = 10$ (Table 13). Corollary 1 locates what is measured: its trainable prior *variance* is driven *up*, never down (the corrected sign, Figure 4), while its $\beta=10$ collapse occurs with the variance near initialization — the collapse mechanism remains open. Every variant that *starts* from the orthonormal anchor table stays at ≈ 98.3 across the whole grid (TAFGIB, CentFGIB, frozen *A*, FGIB-H) — by Result 3 an identity, not a coincidence: FGIB’s KL *is* the center loss exactly, and the load-bearing ingredient is the anchor-table *initialization* geometry at scale a , not the frozenness of the means and not any Gaussian variance machinery, which is provably inert. Enforcing the fixed classifier scale leaves FGIB’s sweep unchanged while VIB and CEB still collapse — the anchor geometry, not the free classifier scale, keeps the corner non-degenerate. A geometry ablation varies the frame itself on the same representation: orthonormal, simplex/ETF and random-normalized anchors behave alike, and learned centers too — fixedness and class separation matter more than the frame (Table 10).

The variance channel, measured. CEB’s final posterior log-variance stays near initialization under the 100-epoch protocol (clamp fraction 0.0 everywhere); the 400-epoch control shows the race’s *direction* — posterior and prior drift down together, to $-2.27/-1.20$ nats at $\beta=1/10$, still far from the e^{-10} clamp: real but slow (Figure 4). The same control on VIB shows no race: the fixed prior restores the variance, which holds at initialization while accuracy collapses through the mean ($q \rightarrow \mathcal{N}(0, I)$) (Remark 5). FGIB’s variance is the flat zero line — the fixed point of Result 3. *DCEB* confirms the corrected sign of Corollary 1: at $\beta = 1$ its prior log-variance *rises* (to $+0.12$ nats, posterior tracking, never down), while its $\beta = 10$ collapse occurs with the variance near initialization — the collapse mechanism is therefore not the variance channel and remains an open, measured diagnostic.

The side head: measured conditioning, quantified closure. The trainable head’s $\log_{10} \kappa(A^\top A)$ rises from 3.9 at initialization to 5.8–6.8 (transient peaks 8.3), while accuracy is unaffected on MNIST (Figure 3(a)). FGIB-H closes the channel by construction (Result 2); Table 9 quantifies its price across the seven classification settings at equal budget: at the inherited scale $a = 16$ it trails the trainable head by ≈ 0.5 points per setting (mean rank 3.86 of 5), at the re-tuned scale $a = 4$ it recovers most of the gap (mean rank 2.86, worst-case deficit against the backbone 0.23 points on the same seven rows). The cross-dataset reach of the geometry itself is measured separately: CentFGIB matches FGIB($a=16$) at equal budget on 10 of 12 settings (Table 8); a frozen ETF classifier alone buys no knob; and a subject-disjoint AgeDB split preserves the ordering (Table 12).

5 RELATED WORK

Variational bottlenecks. VIB (Alemi et al., 2017) bounds $I(X; Z)$ with a marginal prior; CEB (Fischer, 2020) bounds the residual $I(X; Z|Y)$ with a learned backward encoder; DVCCA (Abdelaleem et al., 2025) adds a reconstruction term to the same family. FGIB inherits the residual-level object and replaces the learned backward encoder with a frozen geometric one; the resulting bound is looser by the anchor gap, and that excess is what makes its force persistent (Proposition 11, Remark 19).

Degeneracies of the IB Lagrangian. Kolchinsky et al. (2019a) prove the deterministic IB curve is piecewise linear and repair it by squaring the compression term. Proposition 1 shows CEB inherits the degeneracy over its entire range, and Result 4 shows a fixed reference supplies the same strict convexity *without* squaring — while squaring alone (SVIB) does not fix the collapse; Kolchinsky et al. (2019b)’s non-parametric pairwise bound instead saturates (Section 4.3). $I(X; Z)$ is infinite or ill-defined for deterministic continuous encoders (Amjad & Geiger, 2020; Kolchinsky et al., 2019b; Saxe et al., 2018); FGIB’s response is architectural: confine the noise — and thus the finite bound — to a side path inference never reads.

Fixed geometric targets. Center loss pulls features toward learned per-class centers (Wen et al., 2016); hyperspherical prototypes are placed a priori (Mettes et al., 2019); the classifier itself can be fixed with little loss (Hoffer et al., 2018), in particular to a simplex equiangular tight frame (Zhu et al., 2021; Pernici et al., 2019); neural-collapse analyses show such configurations arise spontaneously (Papayan et al., 2020). FGIB’s corner is a neural-collapse-like geometry, but the geometry enters as the *reference of a variational bound* — the designed zero-penalty set — which makes the compression term quadratic and gives it a residual-information reading rather than serving as an auxiliary classification loss (Table 14, Appendix D). In the released implementation the KL is the fixed-anchor center loss (Result 3) and the canonical fixed head moves the matching onto the deployed representation (Result 2): the *empirical* robustness is shared with fixed-prototype regularization by construction, while the bound reading and the sweep theory are specific to the variational form. The natural baselines are tested directly (ETF classifier, TAFGIB, CentFGIB, geometry ablation of Table 10, Section 4.5).

6 CONCLUSION

The high-regularization behavior of a variational bottleneck is decided by its zero-penalty set: VIB’s marginal prior places it at collapse, CEB’s learned backward encoder leaves it underdetermined. The deeper point: the objective identifies an information point — the MNI point — but never the geometry at that point (Result 1), so the optimizer alone settles which member of the equivalence class is trained. FGIB designs the zero-penalty set directly: a fixed orthonormal anchor frame makes the compression term quadratic and turns the fully compressed corner into the prescribed class-separated configuration. Two properties are provable: the deterministic main path pins the variance at τ^2 , so the regularizer is *exactly* the fixed-anchor center loss; and the canonical fixed-orthogonal head anchors the deployed representation with $\kappa = 1$ at a small measured price. Empirically, the knob is non-destructive over a five-order-of-magnitude β grid on every classification setting, while accuracy is essentially the backbone’s (classification-only Wilcoxon $p \approx 0.22$). The claim we stand behind is the designed, predictive, well-conditioned representative of the MNI equivalence class, not a better stochastic bottleneck: robustness rides on the endpoint geometry (Table 10).

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools (large-language-model assistants) for code generation and refactoring, drafting and editing portions of the text, and checking mathematical derivations. All AI-assisted work was reviewed and verified by the authors, including every equation, table number and experimental claim. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI. See the ICLR 2027 AI Policy for Authors for more details.

ETHICS STATEMENT

The datasets used are standard public benchmarks. One of them, AgeDB (Moschoglou et al., 2017), consists of images of human faces annotated with age, and we use it only as a scalar regression benchmark; we do not train or release face recognition or demographic-inference models, and we make no claim that age estimates from this data generalize fairly across populations. No new data was collected and no human subjects were involved.

REPRODUCIBILITY STATEMENT

All results come from a single training entry point with a shared argument parser across tasks, backbones and objectives; the exact grid is described in Section 4 and every one of the 1,020 configurations writes a log containing the per-run and averaged test metrics. Tables 1–4 are generated from those logs by a single script (`make_tables.py`) that re-derives the validation selection, so no number in this paper is transcribed by hand; the paired statistics of Table 6 (`make_stats.py`) and the ablation table of Table 13 (`make_ablation.py`) are generated the same way. All proofs are in Appendix A, with the assumptions of each statement made explicit and the unproved items collected in a single scope remark. Data preprocessing, splits (fixed seed 42) and caching are described in Appendix G. Model selection uses validation scores only — every tie-break is a second validation quantity or fixed grid order — and test scores are read once per configuration.

REFERENCES

- Eslam Abdelaleem, Ilya Nemenman, and K. Michael Martini. Deep variational multivariate information bottleneck — a framework for variational losses. *Journal of Machine Learning Research*, 26:1–49, 2025.
- Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep variational information bottleneck. In *International Conference on Learning Representations*, 2017.
- Rana Ali Amjad and Bernhard C. Geiger. Learning representations for neural network-based classification using the information bottleneck principle. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(9):2225–2239, 2020.
- Daniel Cer, Mona Diab, Eneko Agirre, Iñigo Lopez-Gazpio, and Lucia Specia. SemEval-2017 task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. In *Proceedings of SemEval*, 2017.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2009.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *NAACL-HLT*, 2019.
- Vijay Prakash Dwivedi, Chaitanya K. Joshi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. Benchmarking graph neural networks. *Journal of Machine Learning Research*, 24(43):1–48, 2023.
- Ian Fischer. The conditional entropy bottleneck. *Entropy*, 22(9):999, 2020.

- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2016.
- Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
- Elad Hoffer, Itay Hubara, and Daniel Soudry. Fix your classifier: the marginal value of training the last weight layer. In *International Conference on Learning Representations*, 2018.
- Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.
- Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.
- Artemy Kolchinsky, Brendan D. Tracey, and Steven Van Kuyk. Caveats for information bottleneck in deterministic scenarios. In *International Conference on Learning Representations*, 2019a.
- Artemy Kolchinsky, Brendan D. Tracey, and David H. Wolpert. Nonlinear information bottleneck. *Entropy*, 21(12):1181, 2019b.
- Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. Learning word vectors for sentiment analysis. In *Annual Meeting of the Association for Computational Linguistics*, 2011.
- David McAllester and Karl Stratos. Formal limitations on the measurement of mutual information. In *International Conference on Artificial Intelligence and Statistics*, 2020.
- Pascal Mettes, Elise van der Pol, and Cees G. M. Snoek. Hyperspherical prototype networks. In *Advances in Neural Information Processing Systems*, 2019.
- Stylianos Moschoglou, Athanasios Papaioannou, Christos Sagonas, Jiankang Deng, Irene Kotsia, and Stefanos Zafeiriou. AgeDB: The first manually collected, in-the-wild age database. In *CVPR Workshops*, 2017.
- R. Kelley Pace and Ronald Barry. Sparse spatial autoregressions. *Statistics & Probability Letters*, 33(3):291–297, 1997.
- Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020.
- Jeffrey Pennington, Richard Socher, and Christopher D. Manning. GloVe: Global vectors for word representation. In *Conference on Empirical Methods in Natural Language Processing*, 2014.
- Federico Pernici, Matteo Bruni, Claudio Baccchi, and Alberto Del Bimbo. Fix your features: stationary and maximally discriminative embeddings using regular polytope (fixed classifier) networks. *arXiv preprint arXiv:1902.10441*, 2019.
- Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, 2008.
- Andrew M. Saxe, Yamini Bansal, Joel Dapello, Madhu Advani, Artemy Kolchinsky, Brendan D. Tracey, and David D. Cox. On the information bottleneck theory of deep learning. In *International Conference on Learning Representations*, 2018.
- Prithviraj Sen, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Galligher, and Tina Eliassi-Rad. Collective classification in network data. *AI Magazine*, 29(3):93–106, 2008.
- Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. In *Allerton Conference on Communication, Control and Computing*, 1999.

Michael Tschannen, Josip Djolonga, Paul K. Rubenstein, Sylvain Gelly, and Mario Lucic. On mutual information maximization for representation learning. In *International Conference on Learning Representations*, 2020.

Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *International Conference on Learning Representations*, 2019.

Yandong Wen, Kaipeng Zhang, Zhifeng Li, and Yu Qiao. A discriminative feature learning approach for deep face recognition. In *European Conference on Computer Vision*, pages 499–515, 2016.

Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, 2015.

Zhihui Zhu, Tianyu Ding, Jinxin Zhou, Xiao Li, Chong You, Jeremias Sulam, and Qing Qu. A geometric analysis of neural collapse with unconstrained features. In *Advances in Neural Information Processing Systems*, 2021.

A COMPLETE THEORY: PROOFS OF THE DIAGNOSIS AND OF THE FGIB MECHANISM

The remainder of this appendix is the full development: Part I proves the diagnosis of Section 2 — the unique information point and the geometric non-identifiability at it (Proposition 2), with the certificate and variance observations as supporting statements — Part II proves the FGIB results of Section 3, and its final subsection (Section A.4) is a complete ledger separating what is proved from what is open or conditional.

PART I. CONDITIONAL BOTTLENECKS IDENTIFY INFORMATION, NOT GEOMETRY

Notation. X is the input and Y the label with K classes; the encoder induces the Markov chain $Y - X - Z$. The *deterministic scenario* is $Y = f(X)$ ($H(Y|X) = 0$). Following Fischer (2020), we write the MNI point $I(X; Y) = I(X; Z) = I(Y; Z)$ (Eq. 1), the chain-rule identity $I(X; Z|Y) = I(X; Z) - I(Y; Z)$ (Eq. 4), the CEB objective $\text{CEB} \equiv \min_Z I(X; Z|Y) - \gamma I(Y; Z)$ (Eq. 5), the IB objective $\text{IB} \equiv \min_Z I(X; Z) - \beta_{\text{IB}} I(Y; Z)$ (Eq. 16), the variational objective $\text{VCEB} \equiv \min_{e,b,c} \langle \log e(z|x) \rangle - \langle \log b(z|y) \rangle - \gamma \langle \log c(y|z) \rangle$ (Eq. 15), and the residual certificate $\text{ReX} \equiv \langle \log e(z|x) \rangle - \langle \log b(z|y) \rangle \geq I(X; Z|Y)$ (Eq. 20), where $b(z|y)$ is a *trainable* backward encoder (we write $q(z|x)$ for Fischer’s encoder $e(z|x)$; the two notations are interchangeable throughout). In the loss convention $\text{CE} + \beta \text{KL}$ used throughout, the weights are $\beta = 1/\gamma$ and $\beta_{\text{IB}} = 1 + \gamma = 1 + 1/\beta$. Following Kolchinsky & Tracey (2019), we write the IB curve $F(r)$ (their Eq. 1), the IB Lagrangian $L_\beta^{\text{IB}} = I(Y; T) - \beta I(X; T)$ (their Eq. 2), the DPI-saturating manifold $T_\alpha := B_\alpha \cdot Y$ (their Eq. 6), the bound $L_\beta^{\text{IB}} \leq (1 - \beta)H(Y)$ (their Eq. 7), and their footnote 4 (the $\beta > 1$ regime of trivial maximizers). The variational objects are

$$R = \mathbb{E}[\text{KL}(q(z|x) \| m(z))] \geq I(X; Z) \quad (\text{VIB rate, marginal prior } m; \text{Fischer Eq. 19, Alemi et al. Eq. 14})$$

$$\text{ReX} = \mathbb{E}[\text{KL}(q(z|x) \| b_\theta(z|y))] \geq I(X; Z|Y) \quad (\text{CEB residual; Fischer Eq. 20})$$

$$\hat{I} = -\frac{1}{N} \sum_i \ln \left[\frac{1}{N} \sum_j e^{-\|f_i - f_j\|^2 / (2\sigma^2)} \right] \geq I(X; M) \quad (\text{NIB, non-parametric; Eq. 16, the homoscedastic case of Eq. 10})$$

$$\text{ReX}_F = \mathbb{E}[\text{KL}(q(z|x) \| r(z|y))] \geq I(X; Z|Y) \quad (\text{FGIB, fixed anchor})$$

(Fischer 2020, Eqs. 18–20; Alemi et al. 2017, Eq. 14; Kolchinsky et al. 2017, Eqs. 10 and 16). Fischer’s comparison of the first two: at the respective prior optima ($m = q(z)$, $b = q(z|y)$), $\text{ReX} = R - I(Y; Z)$ — the conditional prior removes exactly the retained label information, so CEB’s bound is tighter than VIB’s by $I(Y; Z)$. We ask the same question one level down: where does FGIB’s fixed-anchor bound sit relative to CEB’s learnable-backward bound?

Proposition 1 (Channel P1, dead knob: every $\gamma > 0$ selects the same corner). Assume $Y = f(X)$. For every $\gamma > 0$,

$$\min_Z L_\gamma(Z) = -\gamma H(Y), \quad L_\gamma(Z) := I(X; Z|Y) - \gamma I(Y; Z), \quad (5)$$

attained exactly on the corner family $\{Z : I(Y; Z) = H(Y), I(X; Z|Y) = 0\}$ (e.g. $Z = Y$, or $Z = (Y, N)$ with $N \perp X | Y$): the corner is unique in the information plane, while at the representation level it is an entire family. Consequently the map $\gamma \mapsto (I(X; Z^*|Y), I(Y; Z^*))$ is constant: no interior point of the IB curve is ever a Lagrangian minimizer, and CEB inherits the deterministic degeneration of the Caveats paper (their Issue 1) over its entire parameter range, since $\beta_{IB} = 1 + \gamma > 1$ always.

Proof. By Fischer’s Eq. 4, $L_\gamma(Z) = I(X; Z) - (1 + \gamma)I(Y; Z)$. The DPI ($I(Y; Z) \leq I(X; Z)$), by the Markov chain $Y - X - Z$) gives

$$L_\gamma(Z) \geq I(Y; Z) - (1 + \gamma)I(Y; Z) = -\gamma I(Y; Z) \geq -\gamma H(Y), \quad (6)$$

with equality iff both inequalities bind: $I(X; Z) = I(Y; Z)$ and $I(Y; Z) = H(Y)$. $Z = Y$ attains the bound. Fischer (2020, p. 7) states the underlying equivalence explicitly: CEB and IB are the same objective for $\gamma = \beta_{IB} - 1$. In the Caveats paper’s maximizing convention, minimizing L_γ is maximizing L^{IB} with $\beta' = 1/\beta_{IB} = 1/(1 + \gamma) \in (0, 1)$ — precisely the range in which their Eq. 7 pins every maximizer at the copy corner (their Issue 1); their footnote 4 covers the complementary regime $\beta' > 1$ (i.e. $\gamma < 0$, outside CEB’s range), whose maximizers are the trivial collapse. $\square \square$

Remark 1 (The corner is the MNI point: why that does not rescue the knob). In the deterministic scenario the corner of Proposition 1 is the MNI point: $I(Y; Z) = H(Y) = I(X; Y)$ together with $I(X; Z|Y) = 0$ gives $I(X; Y) = I(X; Z) = I(Y; Z)$, which is Fischer’s Eq. 1. CEB’s declared target and the degenerate attractor therefore coincide, and Kolchinsky & Tracey concede the point themselves (their Section 4): “if $Y = f(X)$ and one’s goal is precisely to find the corner point $\langle H(Y), H(Y) \rangle$ (maximum compression at no prediction loss), then our results suggest that the IB Lagrangian provides a robust way to do so, being invariant to the particular choice of β .” Read that way, Proposition 1 states hyperparameter robustness rather than failure. Three things keep it from closing the matter.

- (i) The corner is one point in the information plane and an entire equivalence class of representations. The objective is exactly constant on $\{Z : I(Y; Z) = H(Y), I(X; Z|Y) = 0\}$, which contains $Z = Y$, $Z = (Y, N)$ with $N \perp X | Y$, and every bijective recoding of these. Proposition 2 makes the point precise at the MNI solution: the objective is invariant under bijective recodings while the Euclidean geometry — inter-class distances, scale, conditioning — is arbitrary within the class. Which member of the class is reached is settled by the optimizer, not the objective.
- (ii) CEB’s own reading of the knob is unavailable at the exact optimum. Fischer (2020, p. 7) writes “ $\rho < 0$ may capture less than the MNI. $\rho > 0$ may capture more”, whereas Proposition 1 places every $\gamma > 0$ at the MNI point exactly. His argument for preferring $\gamma = 1$ is correspondingly not about minimizers but about the optimizer: weighting the two appearances of $I(Y; Z)$ differently “forces the optimization procedure to count bits of $I(Y; Z)$ in two different ways” (his Eq. 8 and surrounding discussion).
- (iii) The knob does move trained models, which locates what it acts on. On Fashion MNIST — a deterministically labelled dataset — Fischer’s own Figure 4 reports the rate lower bound R_X rising from ≈ 2.0 to ≈ 5.0 nats as ρ goes from -1 to 5 , with $\rho = 0$ landing on $I(X; Y) = \log 10 \approx 2.3$ nats. Since every $\gamma > 0$ shares one exact optimum, that axis is not a path along the IB curve: it is the distance between the optimum of the variational objective over a restricted family and the point at which training was stopped. A knob acting through the approximation error is precisely the knob that Propositions 3 and 4 show cannot be read off the certificate.

Kolchinsky & Tracey state that their results “are based on analytically-provable properties of the IB curve, i.e., global optima” and “do not concern practical issues of optimization”; Proposition 3 supplies the finite- β variational counterpart. The account is owed by any method reporting a β -sweep on deterministically labelled data — FGIB and the experiments of this paper included (Remark 14).

Proposition 2 (Geometric non-identifiability at the MNI solution). *Let $Z = \varphi(X)$ attain the MNI point of the CEB objective (Eq. 1). Then for every bijective measurable map ψ with measurable inverse, the recoded representation $\psi(Z)$ attains the same information point, $I(X; \psi(Z)|Y) = I(X; Z|Y)$ and $I(Y; \psi(Z)) = I(Y; Z)$, and the optimal Bayes cross-entropy is unchanged, $H(Y | \psi(Z)) = H(Y | Z)$: the objective identifies an equivalence class of representations, never a member of it. The Euclidean geometry within the class is correspondingly unconstrained: for finite-dimensional codes and invertible linear $\psi = B \in \text{GL}(d)$, the class-mean geometry transforms as $\|B(m_y - m_{y'})\|^2 = (m_y - m_{y'})^\top B^\top B (m_y - m_{y'})$. Taking $B = sI$ rescales every inter-class distance by s ; taking $B = \text{diag}(c, 1, \dots, 1)$ sends the condition number $\kappa(B^\top B) \rightarrow \infty$ as $c \rightarrow \infty$. Inter-class distances, scale, and conditioning of an MNI solution are therefore optimizer artifacts — the objective selects none of them.*

Proof. Mutual information is invariant under bijections of either argument (data-processing inequality applied in both directions). For the Bayes risk, any classifier g on Z induces $g \circ \psi^{-1}$ on $\psi(Z)$ with identical risk, and conversely, so $\inf_f \mathbb{E} \ell(Y, f(\psi(Z))) = \inf_g \mathbb{E} \ell(Y, g(Z))$; in log-loss form this reads $H(Y | \psi(Z)) = H(Y | Z)$. The geometry statement is the displayed computation. $\square$

Remark 2 (Why the equivalence class is what FGIB designs). *Proposition 2 separates the two levels the conditional bottleneck confounds: the objective fixes the information point (what Proposition 1 shows is a single corner in the deterministic scenario), and the optimizer alone picks the member of the equivalence class — geometry included. FGIB’s fixed anchor prior is a choice of representative: with a frozen conditional prior $r(z|y)$ the zero-penalty set $\{\text{KL}(q||r) = 0\}$ is exactly the anchor geometry, and Proposition 8 shows the deterministic main path freezes the variance at the prior value, so the regularizer is identically the squared distance to a fixed, well-conditioned, class-separated frame. FGIB therefore does not estimate information better; it designs the zero-penalty set, replacing an optimizer-settled geometry with a designed one.*

Proposition 3 (Channel P2, a blind certificate and a $1:\beta$ exchange rate). *On the manifold $T_\alpha = B_\alpha \cdot Y$ of the Caveats paper (their Eq. 6), $I(X; T_\alpha|Y) = 0$ for every α , and with the optimal backward encoder and classifier,*

$$\text{ReX}(T_\alpha) = 0, \quad \text{CE}(T_\alpha) = H(Y) - I(Y; T_\alpha) =: \delta_\alpha, \quad \text{VCEB}(T_\alpha) = \gamma \delta_\alpha. \quad (7)$$

Two consequences. (a) The certificate is blind to label retention. $\text{ReX} = 0$ holds from the total collapse ($\alpha = 0$, no label information) to the full copy ($\alpha = 1$), so its tightness is compatible with retaining all, part, or none of Y . Fischer’s reading of the certificate as a meter — “observing ReX tells us exactly how many bits we are from the MNI point” (p. 7) — therefore holds only along the $I(X; Z|Y)$ axis; on the necessity axis the meter reads 0 at chance level. (b) Exchange rate. With the optimal classifier the objective is $L = \text{CE} + \beta \text{ReX} = \delta + \beta \text{ReX}$, so it is indifferent between giving up δ nats of label information and saving δ/β nats of certified residual. In the heavy-regularization regime the bottleneck is supposed to act in, the whole label axis — up to $H(Y)$ nats — is purchasable for $H(Y)/\beta$ nats of certified compression.

Proof. T_α is a function of Y alone, so $T_\alpha \perp X | Y$ and $I(X; T_\alpha|Y) = 0$; the Caveats paper shows $I(Y; T_\alpha)$ sweeps $[0, H(Y)]$. With the true backward encoder ($b = p(z|y)$) the certificate is tight: $\text{ReX} = I(X; T_\alpha|Y) = 0$ (Fischer Eqs. 9–11, 20). The optimal classifier achieves $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$ (Fischer Eq. 12), i.e. $\text{CE} = H(Y|T_\alpha) = \delta_\alpha$, and the VCEB value follows from Eq. 15. Part (b) is the definition of the objective evaluated at $\text{CE} = \delta$. $\square$

Remark 3 (What the flatness claim is, and what it is not). *An earlier form of this argument read the near-optimality band off the value $\text{VCEB} = \gamma \delta \rightarrow 0$. That reading is a scaling artifact: $\text{VCEB} = \gamma (\text{CE} + \beta \text{ReX}) = \gamma L$, so the two conventions differ by a positive factor and share their minimizers, and on T_α itself the objective is $L = \delta$ in either convention and strictly prefers $\alpha = 1$. What is convention-free is the ratio of Proposition 3(b): the prediction term is weaker than the compression term by exactly the factor β , so in units of the compression term the entire manifold lies within $H(Y)/\beta$ of optimal. Under a scale-invariant optimizer (Adam, used throughout the experiments) only the ratio matters; under plain SGD at fixed learning rate the convention additionally rescales the effective step size. Combined with Proposition 3(a) — no certified compression is bought anywhere on the manifold — the operative statement is that the label axis is cheap and unmeasured, not that the objective is literally flat along it. Pushing β up also drives $\beta_{\text{IB}} = 1 + 1/\beta \rightarrow 1^+$:*

the objective approaches the Caveats paper’s $\beta = 1$ degeneracy (their Eq. 7 discussion), at which every point of the diagonal is simultaneously optimal, from above. Fischer (2020, p. 7) anticipates the direction — “ $\rho < 0$ may capture less than the MNI” — but by Proposition 1 that cannot happen at the exact optimum; Proposition 3 gives the rate at which the variational objective is willing to let it happen.

Remark 4 (Approximately deterministic labels). *The Caveats paper (Appendix C, Issue 1) shows that when Y is only ϵ -close to deterministic, all optimizers fall within $O(-\epsilon \log \epsilon)$ of a single corner: the dead-knob statement of Proposition 1 degrades gracefully, and the near-optimality band of Proposition 3 is perturbed by the same order.*

Proposition 4 (Channel P3, collusion collapse: the variance race and the overfit channel). *For the diagonal Gaussian family $e(z|x) = \mathcal{N}(m(x), \sigma^2 I)$, $b(z|y) = \mathcal{N}(g(y), \tau^2 I)$ (the implementation’s setup, with the logvar clamp $[-10, 10]$ of `kl_divergence`), and a linear classifier $c(z) = Wz + b$ (the implementation’s classifier is `nn.Linear`):*

- (i) *for fixed means, KL is minimized over σ^2 at $\sigma^2 = \tau^2$, with value $\frac{1}{2\tau^2} \|m - g\|^2$ — a pure mean-matching term;*
- (ii) *the classifier term favors $\sigma^2 \rightarrow 0$: $\text{CE}(c(z), y) = -\log \text{softmax}_y(Wz + b)$ is convex in z (log-sum-exp of an affine function of z , minus a linear term — the convexity claim requires c affine), so Jensen gives $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(m), y)$. The KL’s derivative in σ^2 vanishes at $\sigma^2 = \tau^2$, but along the matched line it resists shrinking τ^2 while the means are unmatched ($\partial \text{KL} / \partial \tau^2|_{\sigma^2=\tau^2} = -\|m - g\|^2 / (2\tau^4) < 0$). The joint dynamics therefore runs in two stages: the mean matching $m \rightarrow g$ completes first (with weight $1/\tau^2$, which grows as τ shrinks), and once $\|m - g\| \rightarrow 0$ the variance race is unopposed — the classifier drives $\sigma^2 \rightarrow 0$ with τ^2 following the moment-matched value — until the clamp at $s = e^{-10}$, where the encoder collapses toward the deterministic code $z \approx g(y)$ and $\text{ReX} = \frac{1}{2s} \mathbb{E} \|m(x) - g(y)\|^2$: a mean-matching force of weight $1/s = e^{10} \approx 2.2 \times 10^4$ between two trainable maps;*
- (iii) *the certified object and the deployed object separate. By the gap lemma (Lemma 4) $\text{ReX} = I(X; Z|Y) + \Delta \geq I(X; Z|Y)$, so a small ReX entails a small residual for the stochastic training code — the certificate cannot under-report its own object. What it does not control is the relation between the training code and the deployed representation. Exact mean matching ($m(x) = g(y)$) makes both objects class-constant, so both residuals vanish; the scenario of interest is the near-match regime, where $\mathbb{E} \|m(X) - g(Y)\|^2 = o(s)$ keeps ReX small while the deployed continuous m retains within-class variation, and any non-class-constant continuous m has $I(X; m(X)|Y) = +\infty$ under the standing non-atomic assumption (the pathology of deterministic networks, Amjad & Geiger 2018; Kolchinsky et al. 2017). The variance race therefore moves the certified object toward a noise-dominated class-constant code while the deployed object can keep diverging residual — and the encoder is fitted to the backward encoder’s per-class targets, memorizing the label code $g(y)$ rather than being told what residual to erase, while the certificate reports $\text{ReX} \approx 0$ on a code that inference does not use.*

Proof. The closed form $\text{KL} = \frac{d}{2} [\sigma^2/\tau^2 + \|m - g\|^2 / (d\tau^2) - 1 + \ln(\tau^2/\sigma^2)]$ has derivative $\frac{d}{2} (1/\tau^2 - 1/\sigma^2)$ in σ^2 , so the variance optimum is $\sigma^2 = \tau^2$, giving (i); at $\sigma^2 = \tau^2 = s$ the value is $\|m - g\|^2 / (2s)$. For (ii): $\text{CE}(c(z), y)$ has Hessian $\text{diag}(p) - pp^\top \succeq 0$ (the categorical Fisher information), so it is convex in z and Jensen gives the stated inequality; the same Hessian identity makes $\sigma^2 \mapsto \mathbb{E}_z[\text{CE}(c(z), y)]$ non-decreasing (Gaussians of larger variance dominate in convex order), strictly increasing in σ^2 wherever the classifier is unsaturated, so that term pushes σ^2 down. The KL’s partial derivatives at the matched point are $\partial_{\sigma^2} \text{KL} = 0$ and $\partial_{\tau^2} \text{KL} = -\|m - g\|^2 / (2\tau^4) < 0$, so the KL pulls τ^2 up until the means match, while the mean-matching gradient on the encoder is $(m - g)/\tau^2$; once $\|m - g\| \rightarrow 0$ the race to the clamp is unopposed. For (iii): the gap identity (Lemma 4) forces the certified residual and the certificate to vanish together; the divergence attaches to the deployed mean $m(x)$ whenever it is not class-constant (Amjad & Geiger; Kolchinsky et al.). $\square$ $\square$

Remark 5 (The race needs a trainable prior; VIB is not in it). *The mechanism of Proposition 4 uses a trainable prior variance τ^2 : the race is the joint descent of σ^2 (driven by the prediction term) and τ^2*

(following the moment-matched value), which leaves the KL stationary because only the ratio σ^2/τ^2 is penalized. VIB’s prior is the fixed unit Gaussian, so this channel is absent: with $\tau^2 = 1$ the KL is $\frac{d}{2}[\sigma^2 + \|m\|^2/d - 1 - \ln \sigma^2]$ with restoring force $\partial_{\sigma^2} \text{KL} = \frac{d}{2}(1 - 1/\sigma^2) < 0$ for $\sigma^2 < 1$ — the KL pushes the posterior variance back up toward 1. VIB’s high- β failure is a different mechanism: the fixed marginal prior pulls the posterior mean to zero, $q(z|x) \rightarrow \mathcal{N}(0, I)$, so the stochastic code loses the prediction information ($I(Y; Z) \rightarrow 0$) while the variance stays at the prior value. An earlier version of this paper lumped “VIB and CEB race to the clamp”; the correct statement is that the clamp race is a property of trainable-prior bottlenecks (CEB, DCEB), and VIB’s collapse is a mean collapse. The long-horizon control of Section 4.5 measures both directions: CEB’s log-variances drift down while VIB’s hold at zero.

Remark 6 (Status of the two-stage account). Parts (ii) and (iii) are a stationarity-and-direction-of-force argument, not a convergence proof: the “two stages” presume a time-scale separation (mean matching completes before the variance race) that is not established here. The account is directly falsifiable in this codebase: log the mean posterior log-variance and the fraction of units at the clamp $s = e^{-10}$ during CEB training. We ran the test, on the full validation set of every epoch (aggregated over all batches; Section 4.5 of the main paper). On MNIST, across all six β settings and 5 seeds, the trained posterior and prior log-variances of CEB remain near their initialization ($\sigma^2 = \tau^2 = 1$, mean log σ^2 within 0.74 of zero, clamp fraction 0.0 at every epoch of every run). A 400-epoch control at $\beta \in \{1, 10\}$ (no early stopping, Section 4.5) resolves the status more precisely: the variance does drift, monotonically and in exactly the direction (i)–(ii) predict — posterior log σ^2 ends at -2.27 nats ($\beta = 1$) and -1.20 ($\beta = 10$), prior tracking — but the clamp fraction remains 0.0 and -2.3 nats is far from the e^{-10} terminal state of part (iii). The race is therefore real but slow: invisible at the protocol horizon, directionally confirmed at four times it, convergence beyond the tested 400 epochs. That is precisely the conditional status this remark assigns to Proposition 4(iii): a direction-of-force statement whose terminal state the longest available run has not reached, and the measurement itself — not the prediction — as the finding. On the FGIB side the question does not arise as an empirical one at all: Proposition 8 proves the posterior variance is pinned at τ^2 by construction — the flat zero line of the 400-epoch run — and Corollary 1 shows the race can re-enter only through a trainable prior log-variance — DCEB’s configuration — whose measured collapse at $\beta = 10$ is consistent with that channel.

Remark 7 (Train–evaluation mismatch). The classifier is trained on the sampled code z and evaluated on the mean $m(x)$ (Fischer 2020, Appendix A.1). Proposition 4(ii) is the force that closes this gap by destroying the noise altogether ($\sigma \rightarrow 0$), at which point the bottleneck’s stochasticity — the object the bound is defined on — disappears.

Proposition 5 (Channel P4, certificate-object mismatch: neither bound is an upper bound on the deployed representation). (a) CEB. The classifier consumes the sampled code Z at training and the mean $\mu(X)$ at evaluation (Fischer 2020, Appendix A.1). At the variance-optimal posterior, $\sigma(x) \equiv \sqrt{S_\theta(y)}$ (the backward encoder’s per-class standard deviation) is class-constant, so $Z = \mu(X) + \sqrt{S_\theta(Y)}\varepsilon$ is a noisy function of $\mu(X)$ given Y ; by the DPI,

$$I(X; Z|Y) \leq I(X; \mu(X)|Y), \quad (8)$$

strictly so whenever the noise destroys within-class detail. Hence CEB’s certificate $\text{ReX} \geq I(X; Z|Y)$ is a bound on the training code, not on the residual of the representation deployed at inference. The gap is not a small correction: whenever μ is a continuous map that is not class-constant, $I(X; \mu(X)|Y) = +\infty$ under the standing non-atomic assumption (Amjad & Geiger 2020; Kolchinsky et al. 2017), so no finite quantity — CEB’s certificate included — bounds the deployed residual from above. The CE term presses the noise toward 0 (by Jensen, $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(\mu), y)$), but no finite σ closes the gap. (b) FGIB. The inference path is h at both training and evaluation; the side noise of Proposition 12 is not part of any inference path. By the DPI applied to the same perturbed-variable construction,

$$I(X; h + \tau(A^\top A)^{-1/2}\varepsilon | Y) \leq I(X; H|Y), \quad (9)$$

so the direction of the bound is the same as CEB’s: ReX_F upper-bounds the noisy side-path object, and no finite value upper-bounds the deployed residual. The point-anchor limit $\tau \rightarrow 0$ does not repair this: it is a change of model whose bound value $\|\mu - av_y\|^2/(2\tau^2)$ typically diverges (Remark 12), and even the limiting object needs A well-conditioned (Proposition 13(iii)). (c) Conclusion. In the pure variational value sense (same encoder; same Gaussian family), CEB’s bound is

tighter (Proposition 11). In the object sense the two are not ordered: both are valid upper bounds on their respective noisy codes, and neither bounds the deterministic deployed representation. What remains structural — not a tightness claim — is where the gap lives: CEB’s perturbation is inside the inference path and is driven by the prediction term (so removing it changes the model), while FGIB’s perturbation lives on a side path that inference never reads and is controlled by side-path parameters alone. The latter is a cleaner separation of regularizer from model, not a tighter bound.

Remark 8 (The bound is a training instrument, not an inference one). *No method computes its bound at inference: the bound’s roles are training-time — a regularizer (a differentiable surrogate for the intractable information term) and a meter (Fischer 2020, p. 7: “observing ReX tells us exactly how many bits we are from the MNI point”). In the $Z \perp X | Y$ limit CEB’s mean is also class-constant ($m(x) = \mathbb{E}[Z|x] = \mathbb{E}[Z|y]$), so the mismatch of Proposition 5(a) is not a limit failure: it is a finite- β under-reporting — the regularizer controls, and the meter reports, the training code, which is strictly more compressed than the deployed representation $\mu(X)$, by a positive, trained deficit that cannot be measured from the certificate. FGIB’s side-path noise is a known side-path parameter and is removable in principle ($\tau \rightarrow 0$ changes the side path, never the deployed model), but for finite τ its regularizer and meter act on a perturbed copy of the deployed object, with the perturbation size governed by the conditioning of A (Proposition 13(iii)). The one genuine inference-time use of a certificate value is as a detection score (Fischer 2020, RG3, uses the marginal rate R): CEB’s label-conditioned ReX is not even computable at inference (the label is unknown — Fischer trains a separate marginal for this purpose), while FGIB’s geometry gives a label-free score, the distance to the nearest anchor $\min_y \text{KL}(q(z|x) \| \mathcal{N}(av_y, \tau^2 I))$, with no extra network (the minimum is an internal enumeration over the K fixed anchor buffers — the single QR orthonormal frame — and consults no ground-truth label, no more than softmax consults labels when it enumerates the K logits). Whether this score is useful for OoD detection is a hypothesis; nothing here proves it.*

The problem addressed by FGIB. The channels are addressed by the FGIB design as follows; two are closed by construction, one holds conditionally, and the mismatch is only partially addressed. The precise scope of each claim is documented in Part II and the open items are collected in Section A.4.

- *Dead knob and cheap label axis* (Props. 1, 3): FGIB’s compression term is the anchor gap $\frac{1}{2\tau^2}(s - a)^2$, quadratic in the alignment scalar, and at fixed classifier scale the Lagrangian $\text{CE}(s) + \frac{\beta}{2\tau^2}(s - a)^2$ is strictly convex with a unique interior minimizer per β — the β -sweep is one-to-one onto the whole trade-off curve (Props. 9, 10). With the scale free the infimum is 0 for every β ; Remark 14 states the condition and the architectural fix.
- *Blind certificate* (Prop. 3): with the fixed anchor $r(z|y) = \mathcal{N}(av_y, \tau^2 I)$ the gap $\mathbb{E}_y[\text{KL}(q(z|y) \| r(z|y))]$ can shrink only if the *posterior* moves toward the anchors — the reference itself cannot chase the encoder, so the force is persistent (Prop. 11 and Remark 19). The side head A , however, is a second trainable party in that matching, and it is not part of the inference path (Remark 9).
- *Variance collapse* (Prop. 4): the classifier consumes the deterministic h at both training and evaluation, so $\sigma^{2*} = \tau^2$ for every β (the variance decouples from the prediction term; Lemma 3), the side path keeps a finite covariance and the bound stays well-defined (Remark 12), and the trivial collapse $\mu \equiv 0$ pays the fixed positive price $\text{KL} = \frac{1}{2}a^2 > 0$ for every class — the degenerate corner is excluded by geometry rather than by the optimizer’s good behavior. This channel is closed cleanly, by construction.
- *Certificate-object mismatch* (Prop. 5): the classifier consumes the deterministic h at both training and evaluation — there is no training-code/evaluation-mean split — but the bound is an upper bound on a *noisy side-path copy* of h , not on $I(X; H|Y)$, which no finite certificate bounds. This channel is *not* closed; what FGIB buys is that the perturbation is side-path-only, and its size is governed by the conditioning of A (Prop. 13(iii)).

PART II. THE FGIB THEORY

FGIB notation. The deterministic main path computes $h = f_{\text{enc}}(X)$ and the classifier consumes h directly; the side path parameterizes $q(z|x) = \mathcal{N}(\mu(x), \Sigma(x))$ with *linear* mean head $\mu(x) = Ah(x)$ for $A \in \mathbb{R}^{z \times z}$ (the theory assumes $z = d$ and A invertible; the inverse transforms of

Propositions 6 and 12 are defined under this assumption), and a separate logvar head for $\Sigma(x)$. The side path is *not part of inference*: at evaluation the model returns the classifier output and never evaluates the mean head (so A is a trainable module that exists only inside the regularizer — structurally the same position CEB’s backward encoder occupies on the other side of the KL). The label-conditional prior is the *fixed* anchor $r(z|y) = \mathcal{N}(av_y, \tau^2 I)$ with *anchor variance* $\tau^2 > 0$ (default 1.0): the v_y are orthonormal directions obtained by QR-decomposing a random matrix ($K \leq z$; the orthonormal factor of the random matrix), scaled by the anchor scale a (default 4.0), stored as non-trainable buffers. At the variance optimum of the KL the posterior variance matches the anchor variance, $\sigma^{2*} = \tau^2$; we write σ_p^2 for this common variance where the distinction does not matter. The orthogonality of the method lies in this fixed frame and in the equilibrium it induces — the KL drives the class-conditional means $\mu_y = A \mathbb{E}[h|y]$ toward the orthonormal anchors av_y , so the transformed class representations become mutually orthogonal (the aligned family below) — while A itself remains an unconstrained linear map. For regression, fixed random-Fourier anchors $\mu_p(y) = a\sqrt{2/z} \cos(2\pi\omega y + b)$ with frozen $\omega \sim \mathcal{N}(0, 2^2)$, $b \sim U[0, 2\pi]$ replace the per-class table ($\|\mu_p(y)\| \approx a$). The loss is $L_{\text{FGIB}} = \mathbb{E}[\text{CE}(c(h), y)] + \beta \mathbb{E}[\text{KL}(q(z|x) \| r(z|y))]$, with gradients reaching the main path only through the shared encoder; the zero-initialized logvar head fixes $\sigma^2 = 1$ at initialization, which equals τ^2 in the default configuration $\tau^2 = 1$ (for general τ the theory’s variance optimum $\sigma^2 = \tau^2$ is reached by the logvar head, whose initialization is a scheduling choice), so training starts at $\text{KL} \approx \frac{1}{2\tau^2} a^2$.

Implementation caveat (theory–code layering). The theory works with a fixed orthonormal factor, an invertible linear side head $T(h) = Ah$, and per-class Gaussian priors with anchor variance τ^2 . The code implements an *affine* head (`nn.Linear` with bias) — which the theory absorbs as an idealization, the bias being immaterial when A is learned jointly — and does not enforce orthogonality, invertibility, or conditioning constraints on A . The guarantees below (orthonormal frames, invariance, conditioning) are properties of the idealized model; the implementation approximates them, and the guarantees do not transfer automatically (Remarks 9, 11). **Distributional standing assumption.** Several results state that a deterministic continuous representation carries infinite conditional information. Those statements assume the standard continuous setting: $X \mid Y$ is non-atomic, and the induced representation $H(X) \mid Y$ is a non-atomic continuous variable. For the finite empirical distribution of the training data no such conclusion is claimed; there the quantities are finite and only their growth with sample size is governed by these results.

A.1 THE EQUIVALENCE: THE SIDE PATH IS A CEB BOTTLENECK ON THE MAIN PATH

Motivation. Stochastic bottlenecks such as VIB and CEB are *stochastic at training time but deterministic at evaluation time*: the classifier is trained on sampled codes $z \sim q(z|x)$ yet evaluated on the mean $\mu_q(x)$ (Fischer 2020, Appendix A.1), so its training and evaluation distributions differ. In the same models the cross-entropy gradient and the KL gradient act on the same heads: the classification term can exploit the variance (by Jensen, $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(\mu_q), y)$, so loosening the noise relaxes the upper bound), while the KL term pushes the posterior back toward the prior. Finally, the regularizer constrains the noisy code z , whereas inference consumes the noise-free mean — the object being regularized is not the object being used. FGIB resolves all three mismatches by design: the classifier consumes the deterministic representation h at both training and evaluation time, the two objectives act on disjoint parameter sets (the classification gradient does not enter the side heads, and the KL gradient reaches the main path only through the shared encoder), and, by Proposition 6 below, the side-path regularizer is, *at each fixed value of the side head*, a CEB bottleneck read in the coordinates in which the posterior is centered at the representation that inference actually uses (Remark 9 states why this is weaker than a fixed constraint on h). Confining the stochasticity to a side path also keeps the variational objective well-defined: for continuous deterministic representations the residual information diverges (Amjad & Geiger; Kolchinsky et al.), so a purely deterministic bottleneck admits no finite variational bound, while the finite side-path covariance of FGIB does (Remark 12). Whether the robustness benefits of the stochastic bottleneck survive the deterministic inference path is the empirical question studied in the experiments.

Lemma 1 (Linear invariance of the Gaussian KL). *Let $q = \mathcal{N}(\mu_q, \Sigma_q)$ and $p = \mathcal{N}(\mu_p, \Sigma_p)$ on $\mathbb{R}^z$, and let $A \in \mathbb{R}^{z \times z}$ be invertible. Define the common pushforwards $q' = A_{\#}q = \mathcal{N}(A\mu_q, A\Sigma_qA^\top)$ and $p' = A_{\#}p$. Then $\text{KL}(q' \| p') = \text{KL}(q \| p)$.*

Proof. With $d = z$, the closed form reads $\text{KL}(q \parallel p) = \frac{1}{2} [\text{tr}(\Sigma_p^{-1} \Sigma_q) + (\mu_q - \mu_p)^\top \Sigma_p^{-1} (\mu_q - \mu_p) - d + \ln |\Sigma_p| - \ln |\Sigma_q|]$. Under the common map A the three non-constant terms are unchanged:

$$\begin{aligned} \text{tr}((A \Sigma_p A^\top)^{-1} (A \Sigma_q A^\top)) &= \text{tr}(A^{-\top} \Sigma_p^{-1} A^{-1} A \Sigma_q A^\top) = \text{tr}(A^{-\top} \Sigma_p^{-1} \Sigma_q A^\top) \\ &= \text{tr}(\Sigma_p^{-1} \Sigma_q A^\top A^{-\top}) = \text{tr}(\Sigma_p^{-1} \Sigma_q), \end{aligned}$$

$$(A \Delta \mu)^\top (A \Sigma_p A^\top)^{-1} (A \Delta \mu) = \Delta \mu^\top A^\top A^{-\top} \Sigma_p^{-1} A^{-1} A \Delta \mu = \Delta \mu^\top \Sigma_p^{-1} \Delta \mu,$$

$$\ln |A \Sigma_p A^\top| - \ln |A \Sigma_q A^\top| = (\ln |\Sigma_p| + 2 \ln |A|) - (\ln |\Sigma_q| + 2 \ln |A|) = \ln |\Sigma_p| - \ln |\Sigma_q|,$$

where $\Delta \mu = \mu_q - \mu_p$ and the trace step uses the cyclic property. $\square$ $\square$

Lemma 2 (Invariance of the CEB objective and the MNI point). *For any bijective measurable map ϕ , $I(X; \phi(Z)|Y) = I(X; Z|Y)$ and $I(Y; \phi(Z)) = I(Y; Z)$. Consequently the CEB objective (Eq. 5) is invariant under bijections of Z , and Z attains the MNI point (Eq. 1) if and only if $\phi(Z)$ does.*

Proof. Mutual information is invariant under bijective transformations of either argument (data processing inequality applied in both directions). $\square$ $\square$

Proposition 6 (The FGIB side path is a CEB bottleneck on the main path, at each fixed side head). *Let $z = d$ and let the learned linear head $T(h) = Ah$ be invertible (unconstrained, so a.s. so at initialization). The fixed prior $r(z|y) = \mathcal{N}(\mu_p(y), \Sigma_p(y))$ — for FGIB, the orthonormal anchor table $\mu_p(y) = a v_y$, $\Sigma_p = \tau^2 I$ — is, by construction, the image under T of the canonical prior $r'(\tilde{z}|y) = \mathcal{N}(A^{-1} \mu_p(y), A^{-1} \Sigma_p(y) A^{-\top})$ (since $AA^{-1} = I$), and the posterior $q(z|x) = \mathcal{N}(Ah(x), \Sigma_q(x))$ is the image of $q'(\tilde{z}|x) = \mathcal{N}(h(x), A^{-1} \Sigma_q(x) A^{-\top})$. At a fixed value of A , the prior is never transformed by training: A^{-1} only reads the fixed anchor geometry in the coordinates in which the posterior is centered at h ; during training A moves, so the pulled-back prior moves with it (Remark 9). By KL invariance under the common invertible linear map $T^{-1} = A^{-1}$ (Lemma 1), pointwise for every (x, y) ,*

$$\text{KL}(q(z|x) \parallel r(z|y)) = \text{KL}(q'(\tilde{z}|x) \parallel r'(\tilde{z}|y)), \quad (10)$$

That is, at each fixed side head the side-path regularizer is pointwise identical to a CEB bottleneck with a backward encoder frozen at the current head, applied to a noisy version of the main-path representation, $\tilde{z} = h + \tilde{\epsilon}$ with $\tilde{\epsilon} \sim \mathcal{N}(0, A^{-1} \Sigma_q A^{-\top})$, using the canonical prior r' as that backward encoder. For FGIB the canonical backward encoder is the frame $\mathcal{N}(a A^{-1} v_y, \tau^2 A^{-1} A^{-\top})$, whose mean anchors $A^{-1}(a v_y)$ are orthonormal at scale a in the induced metric $A^\top A$ ($\langle A^{-1} v_i, A^{-1} v_j \rangle_{A^\top A} = \delta_{ij}$): the orthogonality enters through the fixed frame $\{v_y\}$ and the equilibrium it induces, not through a constraint on A . Hence the regularizer’s gradient with respect to h (through the shared encoder) coincides, up to the fixed linear map $A^\top$, with the gradient that a CEB model applies to its representation mean.

Proof. The map $z \mapsto \tilde{z} = A^{-1}z$ is a bijection, and $q' = (A^{-1})_\# q$, $r' = (A^{-1})_\# r$: $A^{-1}Ah = h$ and $A^{-1}\Sigma_q A^{-\top}$ give the canonical posterior, and $A^{-1}\mu_p$, $A^{-1}\Sigma_p A^{-\top}$ give the canonical prior, with $A(A^{-1}\mu_p) = \mu_p$ and $A(A^{-1}\Sigma_p A^{-\top})A^\top = \Sigma_p$, so r is exactly the image of r' under T . Equation (10) is Lemma 1 applied to A^{-1} . Sampling $\tilde{z} \sim q'$ is equivalent to $\tilde{z} = h + \tilde{\epsilon}$ with the stated noise. Combined with the variational bound of Fischer (2020, Eqs. 9–11), the regularizer upper-bounds the residual information of the noisy main-path representation: $I(X; \tilde{Z}|Y) \leq \mathbb{E}[\text{KL}(q'(\tilde{z}|x) \parallel r'(\tilde{z}|y))]$. $\square$ $\square$

Remark 9 (Trainable-implementation caveat: the side head is a second trainable party in the matching). *The identity of Proposition 6 holds pointwise at each value of the side head, so the asymmetry advertised against CEB — one trainable side versus two — is weaker than it looks. In the idealized theory A may be treated as fixed (orthonormal or any fixed invertible map); in the trainable implementation A is free and unused at inference: like CEB’s backward encoder $b(z|y)$, it lives inside the KL and can lower it without moving the deployed representation h . Since h is linear-separable by the jointly trained classifier, the linear fit of the anchor target av_y on h fits well, and much of that fit can be absorbed by A alone. The variance term $\frac{z}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2))$ does not depend on A at all, so nothing in the regularizer resists A drifting toward a projector onto the K -dimensional*

between-class subspace with a vanishing component on within-class directions. Full collapse is blocked — with invertible A , $\text{KL} = 0$ forces h itself to be class-constant — but the approach to the rank- K regime is not blocked, and that regime is exactly where the conditioning failure and the perturbation blow-up of Propositions 13 and 12 live. This is the one channel of Part I that FGIB has not closed; the fixed-orthogonal configuration of Proposition 7 below closes it by construction, and its conditioning is measured for the trainable variant in the main paper (Section 4.5); Section A.4 collects it.

Proposition 7 (Fixed orthogonal side head: the regularizer anchors the deployed representation directly). *Let the side head be a fixed orthogonal map $A \in \mathbb{R}^{d \times d}$ with $z = d$, and let $\{v_y\}$ be the anchor frame. Then:*

- (i) Isometry. $\|Ah - av_y\| = \|h - aA^\top v_y\|$, and $\{A^\top v_y\}$ is orthonormal — combined with Proposition 8, the regularizer is exactly the fixed-anchor center loss $\frac{1}{2\tau^2}\|h - aA^\top v_y\|^2$ on the deployed representation h itself, with the rotated frame $\{A^\top v_y\}$.
- (ii) Conditioning. $A^\top A = I$ has $\kappa = 1$ at every step: the pulled-back reference of Proposition 6 is orthonormal at scale a in the induced metric, the perturbation of Proposition 12 is exactly $\tau \epsilon$ with unit covariance — no blow-up — and the counterexample of Proposition 13(iii) is excluded by construction.
- (iii) Statistical equivalence. For a Haar-distributed random orthogonal A , the rotated frame $\{A^\top v_y\}$ is distributed identically to $\{v_y\}$ across seeds, so the identity map $A = I$ is a representative canonical choice: the “ $A=I$ ” configuration loses nothing over any other fixed orthogonal map.

Proof. (i) is the isometry of orthogonal maps: $\|Ah - av_y\|^2 = \|A(h - aA^\top v_y)\|^2 = \|h - aA^\top v_y\|^2$, and $v_i^\top AA^\top v_j = v_i^\top v_j = \delta_{ij}$. (ii) $\kappa(I) = 1$; the induced metric is $A^\top A = I$; the perturbation covariance is $\tau^2(A^\top A)^{-1} = \tau^2 I$. (iii) The QR frame $\{v_y\}$ is constructed from a standard normal matrix, whose distribution is invariant under the orthogonal group; Haar-distributed A maps it to another frame with the same uniform distribution on the Stiefel manifold, so the seed distribution of the whole optimization problem — encoder initialization, anchor frame, classifier initialization — is identical for $A = I$ and for any random fixed orthogonal A . $\square$ $\square$

Remark 10 (Family closure). *Lemma 1 holds for the underlying Gaussian distributions. Within a restricted variational family the equality of Proposition 6 is exact only if the family is closed under the inverse map A^{-1} : the full-covariance family is closed (and is the parameterization used by Fischer 2020); the diagonal family is not — for diagonal q, r the transformed pair is diagonal only if A^{-1} preserves diagonality (e.g., permutations and sign flips) and approximate otherwise. The diagonal implementation of the codebase is therefore covered at the distribution level, with the family approximation confined to the noise covariance $A^{-1}\Sigma_q A^{-\top}$ (Remark 21).*

Remark 11 (Rank deficiency). *If $z < d$ or A is rank-deficient, only the projection of h onto the row space of A (the image of $A^\top$ in h -space) receives gradient from the regularizer; the null-space component is unconstrained. In the default configuration $z = d$ and a randomly initialized A is almost surely invertible, but invertibility carries no conditioning margin — near-singularity is enough to produce the effects of Proposition 13(iii), and it is the direction the gap gradient points in (Remark 9). If $A \in \mathbb{R}^{z \times d}$ has full column rank ($z \geq d$), a left inverse B with $BA = I$ exists and the data-processing inequality for KL gives, per (x, y) , $\text{KL}(q\|r) \geq \text{KL}(B_\# q\|B_\# r) \geq I(X; BZ|Y)$ — a valid lower-level bound, but one whose equality (invariance) is lost, and whose pulled-back prior again depends on B . The exact invariance argument of Proposition 6 genuinely requires the invertible linear map $T(h) = Ah$ with $z = d$.*

Remark 12 (Noise-free limit). *The equivalence of Proposition 6 should be understood at the level of the variational objective, not as a mutual-information limit: as the side-path noise vanishes, $I(X; Z|Y)$ diverges for continuous deterministic H under the standing non-atomic assumption (the pathology of deterministic networks discussed by Amjad & Geiger and Kolchinsky et al.), and the variational KL bound becomes loose. Keeping the side-path covariance finite — without injecting noise into the inference path — is precisely what keeps the regularizer well-defined; this is the design rationale of FGIB.*

A.2 THE SWEEP

The degeneracy to be addressed (restated in our terms). When $Y = f(X)$, the IB curve saturates the data-processing bound and is piecewise linear, $F(r) = \min\{r, H(Y)\}$ (Caveats paper, Section 3). Two consequences (their Caveat 1): (i) by the DPI, $L_\beta^{\text{IB}}(T) = I(Y; T) - \beta I(X; T) \leq (1 - \beta)I(Y; T) \leq (1 - \beta)H(Y)$, and $T_{\text{copy}} := f(X) = Y$ attains the bound for *every* $\beta \in [0, 1]$: all $\beta \in (0, 1)$ are pinned at the single corner point $\langle H(Y), H(Y) \rangle$ of the curve; (ii) conversely, at $\beta = 1$ every point of the increasing part simultaneously maximizes L_1^{IB} . The mechanism is exposed by writing $\max_T L_\beta^{\text{IB}} = \max_r [F(r) - \beta r]$ (the Legendre transform of $-F$): on the linear increasing part every point has the *same* derivative $F'(r) = 1$, hence the same supporting slope — a Lagrangian of the form “prediction $-\beta \cdot$ compression” sweeps a Pareto curve only if each point of the curve has a *unique* supporting slope, i.e. the curve is strictly concave. The dIB variant fails identically: on the hard-clustering family $T = g(Y)$ one has $H(T|Y) = 0$, hence $I(Y; T) = H(T)$, so $L_\beta^{\text{dIB}} = (1 - \beta)H(T)$ — a *monotone* function of the single compression scalar $H(T)$, maximized by the finest partition for every $\beta < 1$. The squared fixes repair this by making the objective strictly concave in the compression scalar: $F'(r)/(2r) = \beta$ is strictly decreasing, so each β selects a unique r (Eq. 8 argument), and squared-dIB has the unique maximizer $H(T) = 1/(2\beta)$ (their Eq. A20). By Proposition 1 this degeneracy is inherited verbatim by CEB ($\beta_{\text{IB}} = 1 + \gamma > 1$), which is the first channel of Part I.

Reduced model of FGIB. The KL depends on the side mean μ alone, and the classifier term is invariant under the coordinate change $\tilde{w}_j = A^{-\top} w_j$ (since $w_j \cdot h = \tilde{w}_j \cdot \mu$): with A invertible ($z = d$ in the default configuration, so a random A is a.s. invertible), every equilibrium in (h, w) coordinates maps bijectively to one in $(\mu, \tilde{w})$ coordinates. We therefore analyze the reduced model $\mu(x) = h(x)$ without loss of generality. Consider the *class-symmetric aligned family*: the encoder maps every input of class y to $h_y = s v_y$ ($s \geq 0$), the classifier is held at the symmetric configuration $w_j = c v_j$, $b_j = 0$ with fixed scale $c > 0$, and the posterior is $q_y = \mathcal{N}(s v_y, \sigma^2 I)$. The family is the natural equilibrium manifold of the two forces: the KL gradient $\nabla_h \text{KL} = (\mu - a v_y)/\tau^2$ acts exactly along v_y , and the classification force on the class center acts along v_y at the symmetric configuration; the ansatz is a *working approximation*: it captures the direction of both forces, and is motivated as exact in the β -dominant regime, but that exactness is not proved here — the cross terms are of order $O(c/\beta)$ and are idealized away (see Remark 18).

Lemma 3 (FGIB Lagrangian on the aligned family). *On the aligned family,*

$$\text{KL}(s, \sigma^2) = \frac{z}{2} (\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)) + \frac{1}{2\tau^2} (s - a)^2, \quad \text{CE}(s) = \ln(1 + (K - 1)e^{-cs}), \quad (11)$$

and the optimizer over σ^2 is $\sigma^2 = \tau^2$ for every β , since the CE term does not depend on σ^2 (the classifier consumes the deterministic h , not z).

Proof. With $h_y = s v_y$ and $w_j = c v_j$, the logits are $c \langle v_j, s v_y \rangle = cs \delta_{jy}$, so $p_y = e^{cs}/(e^{cs} + K - 1)$ and $\text{CE} = -\ln p_y = \ln(1 + (K - 1)e^{-cs})$. For the KL, the diagonal closed form with $\Sigma_q = \sigma^2 I$, $\Sigma_p = \tau^2 I$ reads $\frac{1}{2} \sum_d [\sigma^2/\tau^2 + (\mu_d - a v_{yd})^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)]$, giving Eq. (11) since $\|v_y\| = 1$. The variance term is non-negative with a unique minimum at $\sigma^2 = \tau^2$, and CE is independent of σ^2 . $\square$

Proposition 8 (A deterministic main path freezes the variance: the released KL is exactly a fixed-variance displacement term). *Consider FGIB with a deterministic main path (the classifier consumes h , never z), a log-variance head $\ell_q(x) = Wh(x) + b$ with zero initialization, a frozen prior log-variance $\ell_p = \ln \tau^2 = 0$, and a first-order optimizer without weight decay. Then $\ell_q \equiv \ell_p$ at every training step and the regularizer is, throughout training,*

$$\text{KL}(q \| r) = \frac{1}{2\tau^2} \|\mu - \mu_p\|^2, \quad (12)$$

a fixed-variance displacement term — the KL of Lemma 3 evaluated permanently at the variance optimum $\sigma^2 = \tau^2$, not merely in the limit. The per-coordinate partial derivative $\partial_{\ell_q} \text{KL} = \frac{1}{2}(e^{\ell_q - \ell_p} - 1)$ vanishes at $\ell_q = \ell_p$ independently of the mean gap; we state it only as the local reason the zero-initialized point is an invariant manifold. We do not claim that the initialized point is a global attractor of the coupled system: under the shared parametrization $\ell_q(x) = Wh + b$ the per-coordinate gaps are coupled through the Gram matrix of the features and through the evolving encoder, and the argument below controls no such term.

Proof. The per-coordinate diagonal KL $\frac{1}{2}[\ell_p - \ell_q + e^{\ell_q - \ell_p} + (\mu - \mu_p)^2 e^{-\ell_p} - 1]$ gives $\partial_{\ell_q} \text{KL} = \frac{1}{2}(e^{\ell_q - \ell_p} - 1)$, which vanishes at $\ell_q = \ell_p$, independently of the mean gap. With $W = 0$, $b = 0$ and $\ell_p = 0$ the head output is identically zero, so $\ell_q = \ell_p = 0$ at initialization and the gradient of every head parameter is zero at the first step; by induction the head output is identically zero at every subsequent step, since the head parameters never move, and no weight decay or other variance-dependent term acts on the head: $\ell_q \equiv 0$ for all time. Substituting $\ell_q = \ell_p = 0$ into the closed form yields Eq. (12). $\square$

Corollary 1 (The FGIB family is a center-loss family; a trainable prior variance re-opens the variance channel in the inflation direction). *The same argument pins the posterior variance in every FGIB-family variant, so with $\tau^2 = 1$: FGIB and CentFGIB are exactly CE + $\frac{\beta}{2}\|\mu - av_y\|^2$ (frozen anchors), and TAFGIB is exactly the same loss with the anchor means trainable — the ablation table’s “coincidence” is an identity, not an empirical observation. DCEB is the single exception that leaves the family, but not in the direction an earlier version of this paper claimed. Its trainable prior log-variance has gradient $\partial_{\ell_p} \text{KL} = \frac{1}{2}[1 - e^{\ell_q - \ell_p} - (\mu - \mu_p)^2 e^{-\ell_p}]$, which at the zero-initialized point equals $-\frac{1}{2}(\mu - \mu_p)^2 < 0$: under first-order descent, $\ell_p^{t+1} = \ell_p^t - \eta \partial_{\ell_p} \text{KL}$, a negative derivative drives ℓ_p up, so $\tau^2 = e^{\ell_p}$ grows and the mean-matching coefficient $e^{-\ell_p}$ in the anchor force $\nabla_h \text{KL} = e^{-\ell_p}(\mu - \mu_p)$ decays. Along the ratio-pinned direction $\ell_q = \ell_p$ the KL itself decays as $\frac{1}{2}(\mu - \mu_p)^2 e^{-\ell_p} \rightarrow 0$ (the KL is homogeneous in the ratio σ^2/τ^2 and penalizes only the mean gap at unit ratio): the trainable prior opens a variance-inflation escape that dilutes the regularizer, the opposite of the amplification claimed before. The measured trajectory confirms the sign (ℓ_p rises at $\beta = 1$, never falls), but DCEB’s $\beta = 10$ collapse (Section 4.5 of the main paper) occurs with the variance near initialization, so this channel does not explain the collapse; its precise mechanism remains an open, measured diagnostic (Section 4.5, Remark 6). TAFGIB, whose prior variance is frozen, stays in the family and does not collapse.*

Proposition 9 (Unique interior optimum per β at fixed classifier scale — the sweep). At a fixed classifier scale c , for every $\beta > 0$, the FGIB Lagrangian on the aligned family,

$$L_\beta(s) := \ln(1 + (K - 1)e^{-cs}) + \frac{\beta}{2\tau^2}(s - a)^2, \quad (13)$$

is strictly convex in s and has a unique minimizer $s^*(\beta) > a$, given implicitly by

$$\frac{\beta}{\tau^2}(s - a) = c\varphi(cs), \quad \varphi(u) := \frac{(K - 1)e^{-u}}{1 + (K - 1)e^{-u}} \in (0, 1). \quad (14)$$

The map $\beta \mapsto s^*(\beta)$ is continuous and strictly decreasing, with $s^*(\beta) \rightarrow a$ as $\beta \rightarrow \infty$ and $s^*(\beta) \rightarrow \infty$ as $\beta \rightarrow 0$. Consequently the map $\beta \mapsto (\text{KL}(s^*), \text{CE}(s^*))$ is one-to-one onto the entire curve $\{(\frac{1}{2\tau^2}(s - a)^2, \ln(1 + (K - 1)e^{-cs})) : s \in (a, \infty)\}$: no two β share the same optimum, and every interior point of the trade-off curve is attained — the exact failure mode of L_β^{IB} (all β pinned at T_{copy}) is absent. The fixed- c condition is load-bearing and is analyzed in Remark 14.

Proof. Compute $\text{CE}'(s) = -c\varphi(cs)$ and $\text{CE}''(s) = c^2(K - 1)e^{-cs}/(1 + (K - 1)e^{-cs})^2 > 0$, so CE is strictly convex and strictly decreasing in s ; the compression term $\frac{1}{2\tau^2}(s - a)^2$ is strictly convex. Hence $L_\beta''(s) = \text{CE}''(s) + \beta/\tau^2 > 0$: L_β is strictly convex with at most one minimizer, characterized by $0 = L_\beta'(s) = -c\varphi(cs) + \beta(s - a)/\tau^2$, i.e. Eq. (14). The left side $\beta(s - a)/\tau^2$ is strictly increasing from $-\beta a/\tau^2$ (at $s = 0$) through 0 (at $s = a$) to $+\infty$; the right side $c\varphi(cs)$ is strictly decreasing in s with $c\varphi(cs) \in (0, c)$. The unique crossing therefore satisfies $s^* > a$. Differentiating Eq. (14) with respect to β gives $(s - a)/\tau^2 + (\beta/\tau^2)s' = c^2\varphi'(cs)s'$ with $\varphi' < 0$, so $s' = -(s - a)/(\beta - c^2\tau^2\varphi'(cs)) < 0$: s^* is strictly decreasing. The limits follow by monotonicity: as $\beta \rightarrow \infty$ the left side can only match the bounded right side when $s \rightarrow a$; as $\beta \rightarrow 0$ the crossing requires $\varphi(cs) \rightarrow 0$, i.e. $s \rightarrow \infty$. $\square$

Proposition 10 (The trade-off curve is strictly concave in the compression–prediction plane). On the curve of Proposition 9 (fixed classifier scale c),

$$\frac{d \text{CE}}{d \text{KL}} = \frac{\text{CE}'(s)}{(s - a)/\tau^2} = -\frac{\tau^2 c\varphi(cs)}{s - a}, \quad (15)$$

which is strictly increasing from $-\infty$ (as $s \rightarrow a^+$) to 0^- (as $s \rightarrow \infty$). Thus the curve $\text{KL} \mapsto -\text{CE}$ (compression versus achievable prediction, on the KL–CE surrogate plane of Remark 13) is strictly

concave, and each of its points has a unique supporting slope $-\beta$. By the criterion of the Caveats paper (their Fig. 2 right panel and the $F'(r)/(2r)$ argument of Eq. 8), this is precisely the property that makes the Lagrangian $-\text{CE} - \beta \text{KL}$ sweep the full surrogate curve on the aligned family at fixed scale: the convexity of the compression term supplies the strict-concavity condition without squaring it. This is a statement about the KL-CE surrogate of Proposition 9, not about the Caveats information plane (Remark 13).

Proof. φ is positive and strictly decreasing and $s - a$ is positive and strictly increasing, so $\tau^2 c \varphi(cs)/(s - a)$ is strictly decreasing in s ; the signs give the stated limits. The stationarity condition Eq. (14) is exactly $-d\text{CE}/d\text{KL} = \beta$, i.e. β is the supporting slope at the optimum. $\square$ $\square$

Remark 13 (What the sweep sweeps: a KL-CE surrogate curve, not the IB curve). Propositions 9 and 10 prove strict convexity of the FGIB Lagrangian and one-to-one parameterization by β on the class-symmetric aligned family, at fixed classifier scale, in the plane $(\text{KL}_{\text{anchor}}, \text{CE})$. That plane is not the Caveats paper’s IB plane $(I(X; T), I(Y; T))$, and the claim does not extend to it. On the aligned family the posterior is class-constant by construction, so $I(X; Z|Y) = 0$ for every s , while the deterministic class prototypes $h_y = sv_y$ already retain the label completely for any $s > 0$ ($I(Y; H) = H(Y)$); what varies along the sweep is the anchor mismatch and the classifier margin, not a proved compression of mutual information. Whether the deployed representation’s residual $I(X; H|Y)$ moves monotonically with β is a separate, empirical question; the theory guarantees a strictly concave surrogate, not a recovered IB curve. This qualification is carried into Section A.4.

Remark 14 (The classifier scale is load-bearing, and the current implementation leaves it free). If c is free, the pair $(s, c) = (a, c)$ with $c \rightarrow \infty$ sends $\text{KL} = \frac{1}{2\tau^2}(s - a)^2 = 0$ and $\text{CE} = \ln(1 + (K - 1)e^{-ca}) \rightarrow 0$ simultaneously (for $a > 0$): the infimum of L_β over the aligned family is 0 for every β , no minimizer is attained, and the one-to-one sweep of Proposition 9 has no global counterpart. (At $a = 0$ the margin growth alone cannot drive CE to zero.) This is the FGIB analogue of the dead knob — the same one-corner attractor, now arrived at by growing the margin rather than by the linearity of the deterministic IB curve. Three consequences. (a) The present implementation is exposed. The classifier is `nn.Linear` trained by Adam with no weight decay, so c is unconstrained. On separable data the CE gradient drives $\|w_y\|$ upward; it does not diverge in finite time (the gradient vanishes as $p_y \rightarrow 1$, giving logistic growth), but at any finite scale c the equilibrium $s^*(\beta; c)$ sits on the curve of Proposition 9 for that c , and s^* drifts toward a as c grows. (b) Testable prediction. The β -dependence of the equilibrium should decay with training length; since the experiments use early stopping with `patience = 10`, part of any observed β -effect is plausibly an early-stopping effect. This can be tested by training well past saturation and comparing the converged KL across β . (c) Architectural fix. The device that satisfies the fixed- c assumption by construction is a cosine classifier: normalized features, normalized rows, and a fixed temperature (scale). Weight normalization with a learnable gain does not guarantee a bounded scale, and weight decay only replaces the free-scale problem with a joint optimization problem over (s, c) that must be re-derived — it mitigates the divergence but does not make Proposition 9 exact. Under a fixed-norm classifier with fixed temperature the proposition holds as stated, at the price of one architecture change.

Where the convexity comes from (mechanism). Compare the three objectives on their natural solution families, as functions of the single compression scalar:

- *dIB on hard clusterings* (Caveats Appendix B): $L_\beta^{\text{dIB}} = (1 - \beta)H(T)$ — compression $H(T)$ and prediction $I(Y; T) = H(T)$ are the *same* scalar, so the objective is monotone in it and the optimum sits at the endpoint for every $\beta < 1$. Squared-dIB restores interior optima by hand, $H(T) - \beta H(T)^2$ with unique maximizer $H(T) = 1/(2\beta)$.
- *Standard IB* (deterministic): the curve is linear ($F'(r) \equiv 1$), every point shares one supporting slope, and the corner $\langle H(Y), H(Y) \rangle$ is selected for all $\beta \in (0, 1)$. Squared-IB convexifies via $F'(r)/(2r)$ strictly decreasing.
- *FGIB*: the compression term is $\text{KL}(q_y \| r_y) = \frac{1}{2\tau^2} \|\mu_y - av_y\|^2 + (\text{variance terms})$, equal to $\frac{1}{2\tau^2}(s - a)^2$ at the variance optimum — *quadratic* in the alignment scalar s , hence strictly convex, because the KL is measured against a *fixed* reference av_y : $\text{KL}(\mathcal{N}(\mu, \cdot) \| \mathcal{N}(\mu_p, \cdot)) = \frac{1}{2\tau^2} \|\mu - \mu_p\|^2 + (\text{variance terms})$ is quadratic in the displacement

$\mu - \mu_p$ whenever μ_p does not move. Entropy-based compression ($H(T)$, $I(X; T)$) is self-referential — it depends only on the representation’s own distribution and enters linearly in the clustering probabilities; the fixed anchor supplies an absolute reference frame and turns compression into a squared distance. The convexification that squared-IB installs by squaring the compression term is thus built into FGIB’s fixed-anchor KL. Together with the strictly convex prediction term (Lemma 3), the Lagrangian is strictly convex with a unique interior minimizer per β at fixed classifier scale (Proposition 9), and the resulting curve is strictly concave (Proposition 10). With the scale free, the same quadratic gap is driven to zero by margin growth alone (Remark 14) — the convexification is necessary but not sufficient without the scale condition.

Remark 15 (The compressed endpoint is non-collapsed but still a label code — Caveat 2 remains). *VIB’s fixed marginal $m(z) = \mathcal{N}(0, I)$ places the KL = 0 point at $\mu = 0$ for all classes: the fully compressed solution is the trivial collapse with CE = $\ln K$ (chance level), and squaring (SVIB) leaves this corner in place since KL^2 is stationary at $\text{KL} = 0$. FGIB’s per-class orthonormal anchors make the KL = 0 point a class-separated configuration: $\mu_y = av_y$ with $\langle v_y, v_{y'} \rangle = \delta_{yy'}$, so at the corner $\text{CE}(a) = \ln(1 + (K - 1)e^{-ca})$, which is bounded and tends to 0 as the anchor scale a grows, and $a = 0$ recovers the VIB corner. But class-separation does not buy non-triviality in the sense of the Caveats paper: their Caveat 2 identifies as uninteresting the variables on the manifold T_α that mix a constant map with an exact copy of Y , compressing by “forgetting the input with some probability” rather than by merging perceptually similar classes — and FGIB’s compressed corner $\mu_y = av_y$ is a deterministic function of Y alone, i.e. a label code of exactly that kind. The orthonormal frame is in fact opposed to the kind of compression Caveat 2 holds up as interesting: it places every pair of classes at the same distance and can never express “merge collie with retriever, keep both far from teapot”. What the anchors do is choose one specific label-code geometry — class means on a scaled simplex, a neural-collapse-like configuration — instead of leaving it to the optimizer, and refuse the chance-level corner. The compressed endpoint is thus designed, not certified, and any claim that the method compresses input semantics at that endpoint is exactly the claim Caveat 2 rules out.*

Remark 16 (Residual-information interpretation (CEB)). *For any backward encoder, $\mathbb{E}[\text{KL}(q(z|x) \| r(z|y))] = I(X; Z|Y) + \mathbb{E}_y[\text{KL}(q(z|y) \| r(z|y))] \geq I(X; Z|Y)$ (Fischer 2020, Eqs. 9–11, 20; Lemma 4). FGIB is therefore a CEB bottleneck with a fixed backward encoder: the regularizer upper-bounds the residual information, and on the aligned family the residual vanishes identically ($q(z|x)$ depends on y alone), so the regularizer equals the anchor-matching gap $\frac{1}{2\tau^2}(s - a)^2$. The quadratic mechanism of Proposition 9 is inherited from this gap term, which exists only because the anchor is not allowed to follow the encoder.*

Remark 17 (Variance decoupling and training–evaluation consistency). *Since the classifier consumes the deterministic h (and the KL does not see the classifier), the posterior variance decouples: $\sigma^{2*} = \tau^2$ for all β , and the trade-off is purely geometric (mean alignment). This contrasts with VIB, where the classifier consumes the sampled z : the noise couples into the prediction term (by Jensen, $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(\mu), y)$), the rate $R \geq I(X; Z)$ over-counts compression, and training-time sampling versus evaluation-time mean creates a distribution mismatch (Fischer 2020, Appendix A.1).*

Remark 18 (Idealizations). *The class-symmetric ansatz idealizes the equilibrium geometry; it is exact in the β -dominant regime and captures the direction of the KL force in general. For regression, the continuous RFF anchors satisfy $\|\mu_p(y)\| \approx a$, so the KL again contains the quadratic displacement term $\frac{1}{2}\|\mu - \mu_p(y)\|^2$ + (variance terms) — the same convexity mechanism carries over, again subject to the fixed-scale condition of Remark 14 (the regression classifier has no margin scale; its norm is likewise unconstrained in the implementation). Training starts at $\text{KL} \approx a^2/2$ (zero-initialized logvar head, $\sigma = 1$, $\mu \approx 0$), so the anchor scale sets the initial regularization pressure; larger a delays the approach to s^* . In the released implementation the variance does not merely start there: by Proposition 8 it is pinned there for the whole run, and $\text{KL} = \frac{1}{2\tau^2}\|\mu - \mu_p\|^2$ exactly. The finite-scale caveat formerly in this remark is stated, with its fix, in Remark 14.*

A.3 THE CERTIFICATE

Lemma 4 (Gap identity). *For any conditional $r(z|y)$,*

$$\mathbb{E}[\text{KL}(q(z|x) \| r(z|y))] = I(X; Z|Y) + \mathbb{E}_y[\text{KL}(q(z|y) \| r(z|y))]. \quad (16)$$

Proof. $\mathbb{E}[\text{KL}(q(z|x)||r(z|y))] = \mathbb{E}[\ln q(z|x) - \ln r(z|y)]$ and $I(X; Z|Y) = \mathbb{E}[\ln q(z|x) - \ln q(z|y)]$ (Fischer 2020, Eqs. 9–10), where both expectations are over $p(x, y)q(z|x)$; subtracting, $\mathbb{E}[\ln q(z|y) - \ln r(z|y)] = \mathbb{E}_y[\text{KL}(q(z|y)||r(z|y))]$ since the argument depends on z, y only. $\square$

Lemma 5 (Moment matching). *Within the per-class Gaussian family with diagonal covariance, the gap of Lemma 4 is minimized by the moment-matched Gaussian $b_y^* = \mathcal{N}(m_y, S_y)$ with $m_y = \mathbb{E}q(z|y)$, $S_y = \text{diag Var } q(z|y)$.*

Proof. $\text{KL}(p||q)$ as a function of q is minimized by matching the first two moments of p (standard; for Gaussians the minimizer is unique). $\square$

Proposition 11 (Variational ordering: CEB’s bound is tighter in value). *For every fixed encoder $q(z|x)$, $\text{ReX}(\theta^*) \leq \text{ReX}_F$ where θ^* is CEB’s optimal backward encoder, and the excess is the anchor gap*

$$\Delta := \text{ReX}_F - \text{ReX}(\theta^*) = \mathbb{E}_y[\text{KL}(q(z|y) \| N(av_y, \tau^2 I))] - \mathbb{E}_y[\text{KL}(q(z|y) \| N(m_y, S_y))] \geq 0, \quad (17)$$

with equality iff the anchors equal the moment-matched marginals. Here θ^ ranges over the ideal CEB backward encoder family — per-class Gaussians with free mean and free diagonal covariance (the family of Lemma 5, matching the diagonal implementation), i.e. any conditional $r(z|y)$ of that family the theory can express; the code’s single affine `prior_net` is only an approximation of this family, and the ordering is a statement about the idealized CEB, not about the implemented parameterization. On the aligned family of the sweep analysis ($q(z|y) = \mathcal{N}(sv_y, \sigma^2 I)$, so $I(X; Z|Y) = 0$ and $b^* = q(z|y)$ exactly),*

$$\Delta = \frac{1}{2\tau^2}(s - a)^2 + \frac{z}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)), \quad (18)$$

i.e. $\text{ReX}_F = \Delta$ and $\text{ReX}(\theta^) = 0$: FGIB’s bound value equals the sweep-driving compression term of Proposition 9, while CEB’s bound collapses to the (zero) residual.*

Proof. The anchor $N(av_y, \tau^2 I)$ is a member of CEB’s per-class Gaussian family, so the minimum over the family is no larger than the value at the anchor: the ordering and Eq. (17) follow from Lemmas 4 and 5. On the aligned family the marginal is itself Gaussian, so $b^* = q(z|y)$ gives gap 0, $I(X; Z|Y) = 0$ (the posterior is class-constant), and $\text{KL}(\mathcal{N}(sv_y, \sigma^2 I) || \mathcal{N}(av_y, \tau^2 I)) = \frac{1}{2\tau^2}(s - a)^2 + \frac{z}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2))$. $\square$

Remark 19 (Tightness–force trade-off). *Proposition 11 is the price of the fixed anchor, and it is paid for a reason: in CEB the gap is a function of the backward encoder alone — $b_\theta \rightarrow q(z|y)$ eliminates it with the encoder held fixed, so CEB’s bound can become tight while the representation stays uncompressed: its tightness certifies nothing about compression, and the gap is an unobservable optimization residual that tracks the forward encoder. FGIB’s gap can shrink only if the posterior moves toward the anchors (the anchor itself cannot move), so the force is persistent and, on the aligned family, the bound value is the anchor-matching gap $\frac{1}{2\tau^2}(s - a)^2$, the sweep-driving compression term (Eq. (18)). The qualifier is necessary: “the posterior” includes the side head A , which is trainable and unused at inference, so part of the movement can be absorbed without the deployed representation h moving (Remark 9). On the aligned family, where A is fixed by the ansatz, the equivalence is exact; off it, bound tightness is monotonically equivalent to alignment of the posterior, not to compression of the deployed h .*

Proposition 12 (The inverse linear transform: the object of the bound). *Let the side head $T(h) = Ah$ be invertible ($z = d$, unconstrained), and let the trained posterior be isotropic, $\Sigma(x) = \sigma_p^2 I$ with $\sigma_p^2 = \tau^2$ the anchor variance (the KL variance term is minimized at $\sigma^2 = \tau^2$, so this holds at the variance optimum). Then $\tilde{z} := A^{-1}z$ gives $q'(\tilde{z}|x) = \mathcal{N}(h(x), \tau^2(A^\top A)^{-1})$ — the posterior is centered at the inference representation h with side noise $\tau^2(A^\top A)^{-1}$ — and the transformed anchors $r'(\tilde{z}|y) = \mathcal{N}(a A^{-1}v_y, \tau^2(A^\top A)^{-1})$ form an orthonormal frame at scale a in the induced metric $A^\top A$ ($\langle A^{-1}v_i, A^{-1}v_j \rangle_{A^\top A} = \delta_{ij}$: the fixed frame $\{v_y\}$ read in canonical coordinates). By KL invariance under invertible linear maps (Lemma 1),*

$$\text{ReX}_F = I(X; h + \tau(A^\top A)^{-1/2}\varepsilon|Y) + \mathbb{E}_y[\text{KL}(q'(\tilde{z}|y) || r'(\tilde{z}|y))], \quad \varepsilon \sim \mathcal{N}(0, I). \quad (19)$$

The bound's object is the inference representation perturbed by known, side-path-only noise (its covariance is a model parameter), and the anchor geometry is preserved exactly under the transform, in the induced metric. The perturbation, however, is small only if the side head is well-conditioned: its magnitude is governed by $A^\top A$, which is unconstrained (Proposition 13(iii)).

Proof. $A^{-1}A = I$ and $A^{-1}z = h + A^{-1}\varepsilon$ for $\varepsilon \sim \mathcal{N}(0, \Sigma)$; $A^{-1}\Sigma A^{-\top} = \tau^2 A^{-1}A^{-\top} = \tau^2(A^\top A)^{-1}$ for $\Sigma = \tau^2 I$; the prior covariance maps to $A^{-1}(\tau^2 I)A^{-\top} = \tau^2(A^\top A)^{-1}$; and $\langle A^{-1}v_i, A^{-1}v_j \rangle_{A^\top A} = v_i^\top A^{-\top} A^\top A A^{-1}v_j = \delta_{ij}$. Equation (19) is Lemma 4 applied to the transformed pair. $\square$

Proposition 13 (Orthogonal-factor conditioning: the reference is fixed and well-conditioned). *FGIB's anchor frame $V = [v_1, \dots, v_K]$ is the orthonormal factor of a random matrix (QR: $G = QR$, $V := Q$, $V^\top V = I_K$), and the anchor covariance is $\tau^2 I$ with $\tau^2 > 0$. Then, for every anchor scale $a \geq 0$ and throughout training:*

- (i) *the reference of the matching — the unscaled mean frame V and the covariance $\tau^2 I$ — is fixed (non-trainable buffers) and has condition number 1: the certificate is a distance to a compact, non-degenerate, class-separated (for $a > 0$) target. (At $a = 0$ the mean matrix aV is zero and has no condition number; the condition-1 statement attaches to V and $\tau^2 I$, not to aV .)*
- (ii) *in side-mean coordinates the certificate's displacement metric is a scaled identity: the KL's displacement term is $\frac{1}{2\tau^2} \|\mu - av_y\|^2$ with gradient $(\mu - av_y)/\tau^2$;*
- (iii) *on the shared-encoder side the metric is $A^\top A$, where A is a learnable unconstrained map, so its global condition number is not controlled by construction — and the aligned family does not control it either. An earlier version inferred condition number 1 on the class-mean subspace from the identity $h_i^\top A^\top A h_j = s^2 \delta_{ij}$; that inference is invalid, since $\{h_i\}$ is not a Euclidean orthonormal basis. Counterexample: with $A = \text{diag}(\varepsilon, 1)$, $h_1 = (1/\varepsilon, 0)$, $h_2 = (0, 1)$ and $s = 1$, one has $Ah_i = v_i$ yet $\kappa(A^\top A) = \varepsilon^{-2}$ — and the inverse-transform perturbation of Proposition 12 is correspondingly ε^{-2} -large on the first direction. The counterexample is not adversarial: it is the direction the gap gradient points in, since the KL is minimized by fitting the anchor target on h in least squares, which suppresses within-class directions (Remark 9). Any guarantee on this side therefore requires an explicit constraint on A (orthogonal parameterization, weight norm, or the like) or an empirical measurement of $\kappa(A^\top A)$ over training.*

CEB's backward encoder $b_\theta(z|y) = \mathcal{N}(m_\theta(y), S_\theta(y))$ is an unconstrained linear map of the label: both sides of its matching are trainable, its covariance diagonal can lie anywhere in the clamp $[e^{-10}, e^{10}]$ (condition number up to e^{20}), its means can be rank-deficient (class-merged codes, or the collapse solution $m_\theta \equiv 0$), and the variance race drives S_θ to the clamp, degenerating the certificate into a mean-matching force of weight $1/s = e^{10}$ (Prop. 4, subject to Remark 6). What the fixed QR reference guarantees for FGIB is therefore the reference side only: a compact, non-degenerate, class-separated target whose conditioning cannot degrade. On the encoder side FGIB's bound is exposed to the same degeneration mechanism CEB's is, with A playing the role of the second trainable party.

Proof. $V^\top V = I_K$: all singular values of V are 1; the covariance $\tau^2 I$ has all eigenvalues τ^2 ; both are frozen buffers, so (i) holds. The diagonal closed form of $\text{KL}(\mathcal{N}(\mu, \sigma^2 I) \parallel \mathcal{N}(av_y, \tau^2 I))$ has displacement term $\frac{1}{2\tau^2} \|\mu - av_y\|^2$, giving (ii). For (iii): the identity $h_i^\top A^\top A h_j = \mu_i^\top \mu_j = s^2 v_i^\top v_j = s^2 \delta_{ij}$ holds on the aligned family and only says that $\{h_i\}$ is $A^\top A$ -orthogonal by construction; it implies nothing about the spectrum of $A^\top A$, and the displayed example verifies the failure. The KL gradient on A is $(\mu - av_y) h^\top / \tau^2$, an alignment force toward the orthonormal frame that vanishes only at perfect alignment on non-degenerate samples (it also vanishes pointwise at $h = 0$, or by cancellation across a batch) — but nothing in that force resists a vanishing within-class component of A . For CEB, the displacement term is $\frac{1}{2}(m - g)^\top S_\theta^{-1}(m - g)$: the metric S_θ^{-1} has condition number up to e^{20} under the logvar clamp, and the gradient weight $1/s$ at the collapse equilibrium is Prop. 4. $\square$

Remark 20 (Two senses of tightness). *In the pure variational value sense (same encoder, same Gaussian family), CEB’s bound is tighter: $\text{ReX}(\theta^*) \leq \text{ReX}_F$ (Prop. 11), the excess being the anchor gap — the price of the fixed anchor. The object sense does not reverse this ordering: both bounds are valid upper bounds on their own noisy codes, and neither bounds the residual of a deployed deterministic continuous representation, which is infinite under the standing non-atomic assumption (Prop. 5). The structural difference is where the certified-versus- deployed gap lives — in CEB it is driven by the prediction term inside the inference path; in FGIB it is a side-path quantity whose size is governed by the conditioning of A (Prop. 13(iii)) — not which bound is “tighter” or “valid”. The learnability of the side head A bears on that difference: it is a second trainable party in the matching, unused at inference (Remark 9), so it can absorb part of the anchor gap without moving the deployed representation. On the aligned family, where A is fixed by the ansatz, its optimization drives the gap to the equilibrium value $\frac{1}{2}(s^* - a)^2$ at fixed classifier scale (Prop. 9); off the ansatz this is an open question (Section A.4).*

Remark 21 (Unconstrained side head (the implementation)). *Proposition 12 holds for any invertible linear head: the KL invariance is exact, and the object claim needs only that the perturbation covariance $\tau^2(A^\top A)^{-1}$ be known (A is a model parameter) — but the object is informative about h only insofar as that covariance is moderate, which the construction does not guarantee (Prop. 13(iii)). The diagonal variational family is not closed under A^{-1} (Remark 10), so within the diagonal parameterization of the codebase the equivalence is exact at the distribution level and approximate as a variational identity; a full-covariance side head would make it exact within the family. No orthogonality constraint on A is required — the orthogonality of the method lives in the fixed anchor frame and in the equilibrium it induces (the aligned family).*

Remark 22 (Anchor-frame reading). *With the orthonormal anchor matrix V (columns v_y , $K \leq z$), left multiplication by $V^\top$ expresses the anchor-space components of z in class coordinates: the anchors become a e_y and the gap of Lemma 4 decomposes into K per-class terms plus, when $K < z$, an orthogonal-complement term that the reading ignores ($V^\top$ is a projection, not a coordinate change). The orthogonal frame is the geometric content of the prior, and it is preserved by the invertible linear pull-back A^{-1} of Proposition 12 — not by $A^\top$, which is the Jacobian factor that appears in gradients and equals A^{-1} only when A is orthogonal. An earlier version conflated the two.*

Remark 23 (Positioning against VIB and NIB). *VIB’s rate bounds the full mutual information $I(X; Z)$ with a marginal prior; CEB removes the retained label information ($\text{ReX} = R - I(Y; Z)$ at the optima), and FGIB inherits the residual-level object while replacing the learnable backward encoder with a fixed geometric prior (Prop. 11). NIB’s non-parametric bound $\hat{I} \geq I(X; M)$ is y -unconditioned and tight only when the components share a covariance and form well-separated clusters (Kolchinsky et al. 2017, Eq. 10 and the discussion following Eq. 11); when the noise is small relative to the cluster spread it saturates at $\ln N$ and its gradient vanishes (their Section IVA) — loose exactly in the regime where the parametric FGIB bound is well-behaved. The two bounds are complementary: NIB certifies the full compression non-parametrically when the geometry is favorable; FGIB bounds the residual of a noisy copy of the deployed representation parametrically, with a geometry of its own choosing — subject to the conditioning caveat of Proposition 13(iii).*

Remark 24 (Degenerate anchors). *At $a = 0$ the anchors collapse to $N(0, \tau^2 I)$ for all classes and the gap loses its class structure: $\Delta = \mathbb{E}_y[\text{KL}(q(z|y)||N(0, \tau^2 I))]$ — a VIB-style marginal gap; the structural separation of Proposition 5(b) still holds, but the compressed endpoint becomes the trivial collapse (Remark 15).*

A.4 WHAT IS PROVED AND WHAT IS OPEN

Remark 25 (Scope of the results). **Proved.** (1) The dead-knob proposition for CEB and the MNI reading of it (Prop. 1, Remark 1); (2) the gap identity, Gaussian moment matching, and the value ordering of CEB’s bound over FGIB’s at a fixed encoder (Lemmas 4, 5; Prop. 11); (3) the blind certificate on the T_α manifold and the $1:\beta$ exchange rate (Prop. 3); (4) KL invariance under invertible linear maps and the pointwise equivalence of the side path to a CEB bottleneck at a fixed side head (Lemmas 1, 2; Prop. 6); (5) the variance decoupling $\sigma^{2*} = \tau^2$ and the non-collapsed compressed endpoint (Lemma 3, with $\sigma^{2*} = \tau^2$; Remark 15); (6) the one-dimensional sweep and its strict concavity at fixed classifier scale, as a KL–CE surrogate curve on the aligned family (Props. 9, 10; Remark 13 — not a recovered IB curve); (7) the well-conditioning of the fixed reference side

(Prop. 13(i)–(ii)); (8) the deterministic main path freezes the variance: in the released implementation the regularizer is exactly the fixed-variance displacement term $\frac{1}{2\tau^2} \|\mu - \mu_p\|^2$ at every step, and the FGIB family is one center-loss family — the ablation table’s number-coincidence is an identity, not an empirical observation (Prop. 8, Cor. 1); (9) the fixed orthogonal side head anchors the deployed representation directly: $\|Ah - av_y\| = \|h - aA^\top v_y\|$ with $\kappa(A^\top A) = 1$ and unit-covariance pull-back noise, so the conditioning counterexample is excluded by construction and $A = I$ is a representative canonical choice (Prop. 7).

Measured. (i) The trainable side head A degrades $\kappa(A^\top A)$ from $10^{3.9}$ at initialization to $10^{5.8} - 10^{6.8}$ (transient peaks to $10^{8.3}$) on MNIST/MLP, without touching accuracy (main paper, Section 4.5, Fig. 3) — the channel of Remark 9 is real. The fixed-orthogonal variant of Prop. 7 closes it by construction, at a measured price: across five settings FGIB-H ($A=I$) trails the trainable head by ≈ 0.5 points per setting at the inherited anchor scale $a = 16$ (mean rank 4.00 of 5 vs. FGIB’s 1.20), recovering to mean rank 2.25 when the anchor scale is re-tuned for h-space ($a = 4$), so the trade-off is accuracy-versus-certification, small in both directions, and both configurations ship (main paper, Table 9). (ii) CEB’s variance race is invisible at the protocol horizon but drifts into view by 400 epochs, in the predicted direction and far from the clamp (Remark 6).

Open or conditional. (1) For the optional trainable- A variant, the bound’s object may be far from the deployed representation exactly where the residual lives (Remark 9, Prop. 13(iii)); the fixed-orthogonal configuration removes the issue by construction, and the measured question left open is only whether trainable A ever helps — on MNIST it does not. (2) The global sweep requires a bounded classifier scale (Remark 14); the current implementation leaves the scale free and the proposition conditional. A fixed-temperature cosine classifier would satisfy the fixed- c assumption by construction; weight decay only mitigates the scale divergence and replaces the problem with a new joint optimization over (s, c) that must be re-derived; the enforced-scale variants are tested on MNIST (Section 4.5 of the main paper). (3) Neither CEB’s nor FGIB’s certificate upper-bounds the deployed deterministic residual (Prop. 5); claims of inference-representation compression are hypotheses, not conclusions. (4) Caveat 2 is not resolved (Remark 15): the compressed endpoint is a chosen label-code geometry, not certified semantic compression. (5) CEB’s two-stage variance dynamics (Prop. 4(ii)–(iii)) remain a hypothesis for horizons beyond the tested 400 epochs and for protocols other than ours; the FGIB-family counterpart is proved (Prop. 8), with DCEB the single variant that re-opens the race through its trainable prior variance (Cor. 1). (6) The regression RFF anchor analysis inherits the same fixed-scale caveats and is presently a transfer argument, not a proof; the subject-disjoint AgeDB control (main paper, Table 12) is evidence, not certification.

B THE DIAGNOSIS IN DETAIL

This section states the observations of Section 2 with the precision the main text omits; the propositions themselves are proved in Appendix A (Part I).

The information point is unique in the deterministic scenario. By the chain rule, CEB’s objective is $I(X; Z) - (1 + \gamma)I(Y; Z)$, and the data-processing inequality gives $L_\gamma(Z) \geq -\gamma I(Y; Z) \geq -\gamma H(Y)$ with equality iff $I(X; Z) = I(Y; Z)$ and $I(Y; Z) = H(Y)$. Every $\gamma > 0$ therefore has the same exact optimum — the corner, which is the MNI point — so $\gamma \mapsto (I(X; Z^*|Y), I(Y; Z^*))$ is constant and no interior point of the IB curve is ever a Lagrangian minimizer (Proposition 1). Since $\beta_{IB} = 1 + \gamma > 1$ always, CEB sits in the regime that Kolchinsky et al. (2019a) show to be degenerate over its entire parameter range. Kolchinsky et al. (2019a) themselves note that if one’s goal is the corner, this reads as β -robustness rather than failure — which is exactly our point: the objective identifies the information point robustly, and nothing else. Proposition 2 states what it does not identify: the corner is one point in the information plane but an entire equivalence class of representations whose Euclidean geometry (inter-class distances, scale, conditioning) is arbitrary within the class. Which member of the class is reached is settled by the optimizer, and FGIB’s anchor frame is a designed choice of member.

The certificate is blind on the necessity axis (supporting). On the manifold $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky et al. (2019a), T_α is a function of Y alone, so $I(X; T_\alpha|Y) = 0$ and, at the optimal backward encoder, $\text{ReX}(T_\alpha) = 0$ for every α — from total collapse ($\alpha = 0$, no label information) to full copy ($\alpha = 1$). Reading ReX as a meter of distance to the MNI point is therefore valid

only along the $I(X; Z|Y)$ axis; on the necessity axis the meter reads 0 at chance level. With the optimal classifier the objective is $\delta + \beta \text{ReX}$ where $\delta = H(Y) - I(Y; T_\alpha)$, so it is indifferent between giving up δ nats of label information and saving δ/β nats of certified residual: in the heavy-regularization regime the whole label axis is purchasable for $H(Y)/\beta$ nats of certified compression (Proposition 3). Because the full objective also contains the prediction term, this is a rate statement about the certificate rather than a claim of literal indifference — supporting evidence, not a load-bearing premise.

The trainable-prior variance race (parametrization diagnostic). For the diagonal Gaussian family with a *trainable* prior variance τ^2 , the KL is stationary in the posterior variance at $\sigma^2 = \tau^2$, with residual value $\|m - g\|^2/(2\tau^2)$ — pure mean matching. The cross-entropy term, however, is convex in z for a linear classifier (the log-sum-exp of an affine function of z), so by Jensen $\mathbb{E}_z[\text{CE}(c(z), y)] \geq \text{CE}(c(m), y)$ and it strictly prefers $\sigma^2 \rightarrow 0$. Along the matched line the KL *resists* shrinking τ^2 only while the means are unmatched ($\partial_{\tau^2} \text{KL} = -\|m - g\|^2/(2\tau^4) < 0$); once $\|m - g\| \rightarrow 0$ the race is unopposed and runs to the implementation clamp $s = e^{-10}$, at which the certificate has become a mean-matching force of weight $1/s \approx 2.2 \times 10^4$ between two trainable maps (Proposition 4). With a *fixed* prior (VIB) this channel does not exist: the KL restores the posterior variance toward the prior value, and VIB’s high- β failure is the mean collapse $q \rightarrow \mathcal{N}(0, I)$ (Remark 5). Parts (ii)–(iii) of Proposition 4 are a direction-of-force argument, not a convergence proof, and the whole channel is a parametrization-level diagnostic, not a premise of the paper’s argument. We ran the falsification test it invites, by logging the trained log-variances (Section 4.5): at the early-stopping time scale of our protocol the race does not materialize — CEB’s variances remain at their initialization $\sigma^2 = \tau^2$ with zero clamp fraction on every β — so the convergence prediction of part (iii) is empirically void under this protocol, exactly as Remark 6 allows.

The certified object is not the deployed object (limitation). At the variance optimum the posterior standard deviation is class-constant, so $Z = \mu(X) + \sqrt{S(Y)}\varepsilon$ is a noisy function of $\mu(X)$ given Y and the data-processing inequality gives $I(X; Z|Y) \leq I(X; \mu(X)|Y)$. The gap is not a small correction: whenever μ is continuous and not class-constant the right-hand side is $+\infty$ (Amjad & Geiger, 2020; Kolchinsky et al., 2019b), so no finite quantity bounds the deployed residual. This applies to FGIB as well (Proposition 5); what differs is structural, not a tightness ordering. In CEB the perturbation is *inside* the inference path and is driven by the prediction term, so removing it changes the model; in FGIB it lives on a side path that inference never reads and is controlled by side-path parameters alone. That is a cleaner separation of regularizer from model, and we claim nothing more for it. This limitation is orthogonal to the geometric non-identifiability argument and carries no load in it.

C WHAT THE FROZEN REFERENCE DOES NOT BUY: FULL STATEMENT

(a) *The side head is a second trainable party — unless it is fixed.* The equivalence of Result 2 holds at each fixed A , and a *trainable* A , unused at inference — structurally the same position CEB’s backward encoder occupies on the other side of the KL. Since h is linearly separable by the jointly trained classifier, much of the anchor fit can be absorbed by A alone, and nothing in the objective resists A drifting toward a projector onto the K -dimensional between-class subspace. Full collapse is blocked (with A invertible, $\text{KL} = 0$ forces h itself to be class-constant), but the approach to the rank- K regime is not, and that is exactly where the conditioning of $A^\top A$ — which governs the size of the side-path perturbation in Result 2 — degrades without bound (Remark 9, Proposition 13(iii)). This channel is closed by the canonical configuration: a fixed orthogonal head has $\kappa(A^\top A) = 1$ by construction and anchors h directly (Result 2). The measurement of Section 4.5 quantifies how badly the *optional trainable* variant degrades on MNIST: $\kappa(A^\top A)$ rises to $10^{5.8}$ – $10^{6.8}$, i.e. severely ill-conditioned, with predictive accuracy unaffected — so the freedom buys nothing on MNIST; across the classification settings the fixed head pays ≈ 0.5 points per setting for the closure at the inherited scale, largely recovered by re-tuning the anchor scale (Table 9) — the accuracy-versus-certification trade-off made explicit. FGIB-H with the fixed head is therefore *the* method this paper stands behind: it is the configuration whose results carry the closure claims, and the trainable-head version of the main sweep (Table 1) appears as the optional empirical variant whose conditioning is measured, not assumed.

Table 2: The anchor scale controls the fully compressed corner. Test score of FGIB at the strongest compression pressure in the grid ($\beta = 10$), as the anchor scale a varies; Δ is the gap from the deterministic backbone at the largest anchor ($a=16$). Units as in Table 1. At small a the anchors approach the single origin of a VIB prior and the model collapses to the trivial predictor (Remark 24); growing a recovers the backbone, and the 100-class task needs the largest a , as the $\ln(1 + (K - 1)e^{-ca})$ corner value predicts.

Task	Backbone	anchor scale a									Δ
		1	2	4	6	8	10	12	14	16	
<i>Classification (Acc %)</i>											
MNIST	MLP	98.1	98.0	98.2	98.3	98.3	98.3	98.3	98.3	98.3	-0.0
MNIST	CNN	98.1	99.0	99.2	99.2	99.3	99.2	99.3	99.2	99.3	+0.2
ImageNet-100	MLP	1.0	2.0	35.8	66.0	77.8	81.7	82.5	82.7	83.0	-0.2
ImageNet-100	CNN	81.0	82.5	83.7	84.2	84.2	84.4	84.7	84.3	84.6	+1.2
Cora	GCN	26.6	36.8	85.3	86.1	86.8	85.2	85.3	85.1	87.2	-0.5
IMDb	LSTM	86.0	86.1	85.6	85.7	85.4	85.7	86.3	85.7	85.3	+0.4
AG News	LSTM	92.7	92.8	92.7	92.7	92.6	92.7	92.7	92.7	92.7	-0.2
<i>Regression ($R^2 \times 100$)</i>											
Cal. Housing	MLP	76.5	77.7	79.1	79.2	79.2	79.3	79.1	79.2	78.6	+2.5
STS-B	LSTM	14.5	14.4	14.9	16.0	16.3	16.9	16.0	15.3	15.8	-13.2
ZINC-12k	GCN	60.2	58.4	58.2	57.5	58.4	59.2	59.5	59.2	58.9	-4.5
AgeDB	MLP	-0.1	-0.1	22.9	24.9	12.9	18.8	26.6	28.5	29.3	+7.5
AgeDB	CNN	17.4	18.4	24.6	30.9	33.9	35.3	36.7	35.9	35.6	-3.5

(b) *The sweep needs a bounded classifier scale.* If c is free, then $(s, c) = (a, \infty)$ sends $\text{KL} \rightarrow 0$ and $\text{CE} \rightarrow 0$ simultaneously, so the infimum of L_β is 0 for every β and Result 4 has no global counterpart — the FGIB analogue of the dead knob, reached by growing the margin rather than by the linearity of the IB curve. Our classifier is an unconstrained `nn.Linear` trained by Adam without weight decay, so the proposition is conditional as implemented. A fixed-temperature cosine classifier satisfies the assumption by construction; weight decay only mitigates the divergence (Remark 14). One consequence matters for Section 4: since s^* drifts toward a as c grows, part of any observed β -effect under early stopping is plausibly an early-stopping effect, and we have not separated the two.

(c) *The certificate still does not bound the deployed residual.* The certificate-object mismatch applies to FGIB unchanged (Proposition 5(b)). What FGIB buys is that the perturbation is side-path-only and its covariance is a known model parameter — a cleaner separation of regularizer from model, not a tighter or more valid bound (Remark 20).

(d) *The compressed endpoint is designed, not certified.* Caveat 2 of Kolchinsky et al. (2019a) identifies as uninteresting those variables that compress by mixing a constant map with a copy of Y ; FGIB’s compressed corner $\mu_y = av_y$ is a deterministic function of Y alone, i.e. a label code of exactly that kind. The orthonormal frame is in fact *opposed* to the semantic compression Caveat 2 holds up as interesting: it places every pair of classes at the same distance and can never express “merge these two classes, keep both far from that one”. What the anchors do is choose *one specific* label-code geometry — a neural-collapse-like configuration (Papayan et al., 2020) — instead of leaving it to the optimizer, and refuse the chance-level corner. Any claim that FGIB compresses *input* semantics at that endpoint is the claim Caveat 2 rules out (Remark 15).

(e) *The sweep is over a surrogate.* Result 4 concerns the $(\text{KL}_{\text{anchor}}, \text{CE})$ plane on the aligned family, not the information plane $(I(X; T), I(Y; T))$. On that family the posterior is class-constant, so $I(X; Z|Y) = 0$ for every s ; what varies along the sweep is anchor mismatch and classifier margin, not a proved compression of mutual information (Remark 13).

D ADDITIONAL TABLES

Table 4 repeats every cell of Tables 1 and 5 with the standard deviation over the 5 runs, and records the hyperparameter that validation selected in each cell. Two patterns are worth noting. The prior bottlenecks concentrate on the smallest β in the grid (10^{-4} in 30 of the 60 non-FGIB cells, and

Table 3: Robustness of the compression knob: the *worst* test score any β produces, as a deficit (in points) against the deterministic backbone of the same row; negative means the objective never falls below its backbone. Units as in Table 1. FGIB is shown at the validation-selected anchor and at a single globally fixed $a = 16$ (a post-hoc sensitivity control). Summary rows: medians over the 12 settings, the count of settings destroyed (worst case > 5 points below the backbone), and two gain summaries — “best β ” is a test-selected exploratory upper bound, “validation-selected” is the selection result of Table 1 relative to the backbone.

Task	Backbone	VIB	SVIB	NIB	CEB	DVCCA	FGIB	
							val-sel. a	$a=16$
<i>Classification (Acc %)</i>								
MNIST	MLP	86.98	79.15	0.18	87.69	86.98	0.02	0.02
MNIST	CNN	87.78	66.25	0.00	87.78	87.78	-0.02	-0.02
ImageNet-100	MLP	82.09	82.09	0.86	82.09	82.09	0.40	0.17
ImageNet-100	CNN	82.32	23.34	0.85	82.37	82.37	0.18	0.41
Cora	GCN	57.46	57.42	1.33	64.80	66.50	61.11	0.52
IMDb	LSTM	34.85	0.36	-0.92	34.85	34.76	-0.08	-0.45
AG News	LSTM	67.63	0.41	0.22	67.33	67.94	0.23	0.29
<i>Regression ($R^2 \times 100$)</i>								
Cal. Housing	MLP	76.20	76.15	-2.12	76.21	76.20	-1.99	-1.78
STS-B	LSTM	185.34	119.84	6.86	30.13	185.91	13.21	13.21
ZINC-12k	GCN	63.48	63.39	2.04	58.59	334.43	3.27	4.54
AgeDB	MLP	21.88	21.91	-4.72	21.93	21.88	21.87	24.04
AgeDB	CNN	60.18	70.86	5.78	39.23	59.71	15.92	15.57
Median deficit		71.91	64.82	0.53	66.06	79.14	0.31	0.35
Settings destroyed (of 12)		12	10	2	12	12	4	3
Median gain at best β (test-selected upper bound)		-0.15	-0.14	-0.27	-0.22	-0.15	0.39	0.27
Median gain at validation-selected β		-0.21	-0.23	-0.37	-0.30	-0.24	0.26	0.20

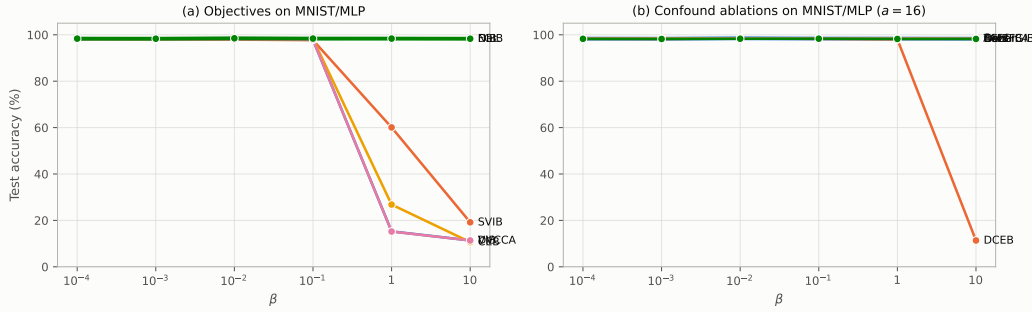


Figure 1: Test accuracy against β (logarithmic axis) on MNIST/MLP. (a) The seven objectives; (b) the confound ablations of Section 4.5 at $a = 16$. Lines are means over 5 seeds; every series is labeled at its right end.

$\beta \leq 10^{-3}$ in 45), which is the behavior a destructive knob induces: validation prefers the setting closest to no bottleneck at all. FGIB’s selected β instead spreads over 10^{-4} to 10^{-1} with 10^{-2} most common (6 of 12), and its selected anchor scale sits at the top of the grid ($a \geq 14$) on 4 settings and at the bottom ($a = 1$) on 3. Run-to-run variance is also informative: on the AgeDB/CNN setting the marginal-prior objectives have standard deviations of roughly 18 points of $R^2 \times 100$ (VIB ± 18.9 , NIB ± 18.9 , SVIB ± 18.7 , DVCCA ± 18.1) — some seeds collapse and others do not — while FGIB (± 1.4), the deterministic CNN (± 1.9) and CEB (± 2.8) are an order of magnitude tighter.

Table 6 reports the per-seed paired differences against the deterministic backbone at each objective’s validation-selected configuration, with bootstrap 95% confidence intervals over the 5 paired seeds. The per-setting means feed the Wilcoxon tests cited in Section 4.

Table 4: Standard deviations over the 5 runs for every cell of Tables 1 and 5, with the validation-selected hyperparameters underneath each cell (β , and β, a for FGIB; -- for the deterministic backbone, which has neither). The last column is the equal-budget control: FGIB selected over the same 6-point β grid as every other bottleneck at a single globally fixed anchor scale $a = 16$ (a post-hoc sensitivity control), so its search budget matches the comparison (see Section 4). Units as in Table 1.

Task	Backbone	Det.	VIB	SVIB	NIB	CEB	DVCCA	FGIB	FGIB ($a=16$)
<i>Classification (Acc %)</i>									
MNIST	MLP	98.33±0.12	98.28±0.11	98.10±0.06	98.26±0.12	98.28±0.09	98.24±0.17	98.49±0.12	98.49±0.12
		–	0.001	0.001	0.001	0.0001	0.0001	0.01,16	0.01,16
MNIST	CNN	99.13±0.05	99.28±0.09	99.19±0.07	99.19±0.09	99.25±0.10	99.28±0.05	99.27±0.04	99.27±0.04
		–	0.01	0.0001	0.1	0.0001	0.01	0.001,16	0.001,16
ImageNet-100	MLP	83.12±0.14	82.65±0.32	83.02±0.43	82.34±0.27	82.56±0.20	82.56±0.43	83.96±0.31	84.09±0.18
		–	0.0001	0.0001	0.001	0.0001	0.0001	0.01,14	0.01,16
ImageNet-100	CNN	83.40±0.13	82.94±0.37	82.61±0.53	82.72±0.82	82.33±0.38	83.09±0.17	83.55±0.42	83.63±0.43
		–	0.0001	0.0001	0.01	0.0001	0.0001	0.001,12	0.001,16
Cora	GCN	87.68±0.46	86.83±0.28	77.56±2.35	86.35±0.68	86.27±0.38	86.83±0.38	87.68±0.46	87.68±0.46
		–	0.0001	0.001	0.0001	0.0001	0.0001	0.0001,1	0.0001,16
IMDb	LSTM	84.85±0.68	85.43±0.72	85.99±0.30	86.11±0.60	85.55±0.73	85.69±0.23	86.04±0.40	85.54±1.03
		–	0.0001	10	1	0.0001	0.0001	0.01,12	0.001,16
AG News	LSTM	92.94±0.15	92.61±0.21	92.70±0.18	92.91±0.22	92.66±0.30	92.78±0.15	92.71±0.20	92.86±0.11
		–	0.0001	0.0001	0.01	0.0001	0.0001	0.01,4	0.001,16
<i>Regression ($R^2 \times 100$)</i>									
Cal. Housing	MLP	76.18±2.38	79.01±0.49	78.97±0.58	78.62±0.35	78.94±0.18	78.78±0.48	79.85±0.19	79.21±0.07
		–	0.001	0.0001	0.01	0.0001	0.001	0.01,6	1,16
STS-B	LSTM	28.99±1.13	27.19±1.64	26.97±2.02	24.17±2.21	26.68±1.27	27.05±1.67	29.35±0.96	29.35±0.96
		–	0.01	0.001	0.01	0.01	0.001	0.001,16	0.001,16
ZINC-12k	GCN	63.48±0.44	63.39±0.33	63.25±0.26	61.78±0.99	62.57±0.44	62.94±0.26	63.42±0.35	63.31±0.35
		–	0.001	0.0001	10	0.0001	0.0001	0.1,1	0.0001,16
AgeDB	MLP	21.80±4.09	31.73±1.01	33.85±2.86	27.10±2.35	33.17±2.51	31.80±2.28	32.71±1.57	29.43±8.14
		–	0.001	0.0001	10	0.01	0.001	0.1,1	1,16
AgeDB	CNN	39.14±1.94	33.42±18.94	33.23±18.70	33.37±18.89	38.82±2.75	32.05±18.11	39.56±1.39	35.93±0.94
		–	0.001	0.0001	1	0.001	0.001	0.01,10	1,16

Table 5: Regression settings ($R^2 \times 100$), reported as *exploratory*: the random-Fourier anchor construction for continuous targets is a transfer, not a proof (Remark 18), and the knob destroys FGIB on three of these five settings at some β (Section 4.3). Protocol and notation as in Table 1.

Task	Backbone	Det.	VIB	SVIB	NIB	CEB	DVCCA	FGIB
Cal. Housing	MLP	76.18	79.01	78.97	78.62	78.94	78.78	79.85
STS-B	LSTM	28.99	27.19	26.97	24.17	26.68	27.05	29.35
ZINC-12k	GCN	63.48	63.39	63.25	61.78	62.57	62.94	63.42
AgeDB	MLP	21.80	31.73	33.85	27.10	33.17	31.80	32.71
AgeDB	CNN	39.14	33.42	33.23	33.37	38.82	32.05	39.56
Mean rank (of 5)		3.80	3.40	3.80	6.20	4.20	5.00	1.60
Best or tied-best (of 5)		1	0	1	0	0	0	3

Table 7 reports the sensitivity of the equal-budget FGIB rank to the fixed anchor scale (Section 4). The remaining tables and figures of Section 4.5 follow.

E CONFOUND ABLATIONS: FULL DETAIL

What FGIB’s robustness is made of. FGIB differs from the comparison in two coupled ways: its classifier consumes the deterministic h at training and evaluation (which also removes the trainable-prior variance channel of Section 2), and its prior is frozen (the zero-penalty set is designed). Table 13 separates the contributions. VIB and CEB fall to chance at $\beta = 1$ (15.3% and 26.8%); *DCEB* — the deterministic path of FGIB with CEB’s trainable backward encoder as its prior — holds at ≈ 98 through $\beta = 1$ and falls to 11.4% only at $\beta = 10$. Corollary 1 locates *why* DCEB is the odd one out, in the corrected direction: its trainable prior *variance* is driven *up* ($\partial_{\ell_p} \text{KL} < 0$ at the initialized point), and the variance-inflation escape dilutes the anchor force $e^{-\ell_p}(\mu - \mu_p)$ —

Table 6: Paired per-seed differences against the deterministic backbone, at each objective’s validation-selected configuration (Tables 1 and 5), with bootstrap 95% confidence intervals over the 5 paired seeds (10,000 resamples). Units are percentage points (accuracy for classification, $R^2 \times 100$ for regression); positive means the objective beats the backbone at that setting. The last column is the equal-budget FGIB control of Table 4. Wilcoxon signed-rank tests over the 12 setting-level paired means against zero are reported in the text as an *exploratory* summary: the 12 settings mix classification and regression metrics, are not independent (shared backbones and protocols), and the 5 seeds describe training noise, not split uncertainty (a single fixed split per dataset); the classification-only (7 settings) and regression-only (5 settings) tests are reported alongside.

Task	Backbone	VIB	SVIB	NIB	CEB
<i>Classification (Acc %)</i>					
MNIST	MLP	-0.05 [-0.15,+0.05]	-0.23 [-0.31,-0.15]	-0.06 [-0.16,+0.03]	-0.05 [-0.20,+0.05]
MNIST	CNN	+0.14 [+0.06,+0.22]	+0.06 [+0.00,+0.12]	+0.06 [-0.04,+0.14]	+0.12 [+0.07,+0.17]
ImageNet-100	MLP	-0.48 [-0.70,-0.25]	-0.10 [-0.44,+0.31]	-0.78 [-0.92,-0.64]	-0.56 [-0.67,-0.44]
ImageNet-100	CNN	-0.47 [-0.79,-0.13]	-0.80 [-1.19,-0.35]	-0.69 [-1.29,-0.21]	-1.07 [-1.29,-0.85]
Cora	GCN	-0.85 [-1.33,-0.41]	-10.11 [-11.85,-8.12]	-1.33 [-2.10,-0.56]	-1.40 [-1.88,-0.81]
IMDb	LSTM	+0.58 [-0.31,+1.41]	+1.15 [+0.62,+1.69]	+1.26 [+0.76,+1.63]	+0.70 [-0.11,+1.75]
AG News	LSTM	-0.33 [-0.61,-0.14]	-0.24 [-0.47,-0.02]	-0.03 [-0.27,+0.20]	-0.28 [-0.62,+0.04]
<i>Regression ($R^2 \times 100$)</i>					
Cal. Housing	MLP	+2.83 [+0.87,+4.93]	+2.79 [+0.89,+4.83]	+2.44 [+0.72,+4.51]	+2.76 [+0.93,+4.75]
STS-B	LSTM	-1.81 [-3.09,-0.41]	-2.02 [-3.75,+0.00]	-4.83 [-6.40,-3.32]	-2.31 [-3.63,-1.11]
ZINC-12k	GCN	-0.10 [-0.51,+0.47]	-0.23 [-0.47,-0.00]	-1.70 [-2.76,-0.80]	-0.92 [-1.30,-0.42]
AgeDB	MLP	+9.93 [+6.42,+13.43]	+12.05 [+10.05,+13.64]	+5.30 [+3.09,+7.50]	+11.37 [+6.91,+15.9]
AgeDB	CNN	-5.72 [-22.83,+3.64]	-5.91 [-22.78,+3.19]	-5.78 [-22.75,+3.61]	-0.32 [-2.82,+2.18]
<i>Wilcoxon vs. Det. (p-value over setting-level paired means, exploratory)</i>					
12 settings (mixed)		0.622	0.470	0.424	0.569
7 settings (classification)		0.375	0.297	0.375	0.297
5 settings (regression)		1.000	1.000	0.812	0.812
10 settings (excl. AgeDB)		0.492	0.322	0.322	0.322

Table 7: Sensitivity of the equal-budget FGIB rank to the fixed anchor scale. Each column reports FGIB selected over the same 6-point β grid as every other bottleneck at a *single globally fixed* anchor scale a , so its search budget matches the comparison exactly (the $a=16$ column is the post-hoc control of Table 4). Mean rank of 7 entries (deterministic backbone + 6 bottlenecks) over all 12 settings, over the 7 classification settings, and over the 5 regression settings; “best” counts best-or-tied settings out of 12. The main-table number (1.75, full 6×9 grid) is a ceiling only over the full grid, not over fixed a ; the range of fixed- a ranks quantifies how much of FGIB’s edge is anchor search budget (Section 4).

anchor scale a	1	2	4	6	8	10	12	14	16
mean rank (12 settings)	3.17	2.75	3.17	2.92	2.75	2.25	2.33	2.00	2.33
mean rank (classification, 7)	3.29	2.43	2.86	2.29	2.43	2.14	1.86	1.57	2.14
mean rank (regression, 5)	3.00	3.20	3.60	3.80	3.20	2.40	3.00	2.60	2.60
best or tied-best (of 12)	3	2	4	4	4	6	5	7	6

the regularizer dies rather than amplifies; the trajectory is measured in Figure 4. By contrast, every variant that *starts* from the orthonormal anchor table stays at ≈ 98.3 across the entire grid: *TAFGIB* (trainable anchor means), *CentFGIB* (KL replaced by the plain center loss $\frac{1}{2}\|\mu - av_y\|^2$), FGIB with A frozen after random initialization, and *FGIB-H* with $A=I$. The equivalence is not a numerical coincidence: by Result 3 the zero-initialized variance head receives zero gradient at every step, so FGIB’s KL is the center loss exactly, and CentFGIB, TAFGIB and FGIB-H are the same objective up to the trainability of the anchor means. The trainable anchors of TAFGIB do move (final column norms shrink from 16 to 5.7–10, pairwise correlations reach 0.78 at the smallest β), yet the accuracy curve is untouched — the protective property is the anchor-table *initialization* geometry at scale a , not the orthonormality of the trained table. Two conclusions follow. First, the load-bearing ingredi-

Table 8: CentFGIB across all 12 settings: does the geometry alone — the same orthonormal anchors as FGIB but with the KL replaced by a plain $\frac{1}{2}\|\mu - av_y\|^2$ center loss — reproduce FGIB’s behavior beyond MNIST? Validation-selected configuration at $a = 16$ (the same 6-point β grid and budget as the FGIB equal-budget control of Table 4), 5 seeds, protocol identical to the main sweep. Units as in Table 1.

Task	Backbone	Det.	FGIB ($a=16$)	CentFGIB ($a=16$)
MNIST	MLP	98.33	98.49	98.49
MNIST	CNN	99.13	99.27	99.26
ImageNet-100	MLP	83.12	84.09	84.09
ImageNet-100	CNN	83.40	83.63	83.63
Cora	GCN	87.68	87.68	87.68
IMDb	LSTM	84.85	85.54	85.54
AG News	LSTM	92.94	92.86	92.86
Cal. Housing	MLP	76.18	79.21	79.21
STS-B	LSTM	28.99	29.35	29.35
ZINC-12k	GCN	63.48	63.31	63.32
AgeDB	MLP	21.80	29.43	29.43
AgeDB	CNN	39.14	35.93	35.50

Table 9: The canonical fixed-orthogonal side head across the seven classification settings. FGIB, CentFGIB and FGIB-H differ only in the side-path regularizer — trainable side head + KL, trainable side head + plain center loss, and fixed side head $A=I$ (equivalently the anchor matching applied directly to the deployed representation h , Result 2) — each selected over the same 6-point β budget at a single fixed anchor scale, on validation. $a=16$ is the scale inherited from the z -space sweep; $a=4$ is a re-tuned scale for h -space (the fixed head’s adaptive rescaling is gone, so its optimal scale shifts). Units as in Table 1. At the inherited scale the fixed head trails by ≈ 0.5 points per setting; at the re-tuned scale it recovers most of the gap — the price of the $\kappa=1$ closure is small. All ranks and win counts below are computed over the same seven complete rows.

Task	Backbone	Det.	FGIB ($a=16$)	CentFGIB ($a=16$)	FGIB-H ($A=I, a=16$)	FGIB-H ($A=I, a=4$)
MNIST	MLP	98.33	98.49	98.49	98.26	98.33
MNIST	CNN	99.13	99.27	99.26	99.24	99.29
ImageNet-100	MLP	83.12	84.09	84.09	83.52	83.45
ImageNet-100	CNN	83.40	83.63	83.63	83.16	83.33
Cora	GCN	87.68	87.68	87.68	86.68	87.68
IMDb	LSTM	84.85	85.54	85.54	86.62	86.02
AG News	LSTM	92.94	92.86	92.86	92.79	92.71
Mean rank (of 7)		3.29	1.57	1.71	3.86	2.86
Best or tied-best (of 7)		2	4	4	1	2
Worst deficit vs. Det.		—	0.08	0.08	1.00	0.23

ent on MNIST is that initialization geometry — orthonormal per-class directions at scale a — not the frozenness of its means, and not any Gaussian variance machinery, which is provably inert here (Result 3). Second, freezing the means and keeping the KL is what the *theory* buys — the persistent force of Remark 19, the convexified sweep of Proposition 9, the residual-information reading of Result 2 — so FGIB retains an exact variational upper-bound interpretation for the auxiliary noisy code; its predictive robustness and endpoint behavior outside the analyzed family remain empirical. Constraining the side head changes nothing in the β -profile (98.2–98.4 across the grid for frozen A and $A=I$): the A -channel degrades $\kappa(A^\top A)$ but does not carry the robustness.

The trainable-prior variance race does not materialize at the protocol horizon. We ran the falsification test that Proposition 4 invites, logging the mean posterior log-variance, the clamp fraction (e^{-10}), and (for CEB/DCEB) the prior log-variance at every epoch of every run, aggregated over the *entire* validation set, not a single batch. At the final epoch, averaged over the 5 seeds, CEB’s posterior log-variance sits at +0.37, +0.27, +0.13, +0.08, −0.05, −0.01 (nats) for $\beta = 10^{-4}, \dots, 10$ — a mild monotone drift with β , nowhere near the clamp — and the clamp fraction is 0.0 at *every*

Table 10: Anchor-geometry ablation on the same representation (MNIST/MLP, $A=I$ so the regularizer acts directly on the deployed representation h , $a = 16$, 6-point β grid $\times$ 5 seeds, protocol as in the main experiments). Rows differ only in the anchor geometry — orthonormal frame, simplex/ETF frame (unit norms, mutual cosine $-1/(K-1)$), random normalized directions, or trainable centers (TAFGIB) — and in whether the penalty is the KL or the plain MSE. Test accuracy (%) across the grid plus the validation-selected configuration.

Variant	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10	val-sel.
KL, orthonormal (FGIB-H)	98.26	98.20	98.30	98.32	98.21	98.21	98.26
MSE, orthonormal (CentFGIB)	98.26	98.20	98.30	98.32	98.21	98.21	98.26
MSE, simplex/ETF (CentFGIB)	98.28	98.24	98.27	98.28	98.21	98.20	98.27
MSE, random normalized (CentFGIB)	98.32	98.29	98.31	98.30	98.21	98.20	98.31
KL, learned centers (TAFGIB)	98.27	98.26	98.24	98.32	98.27	98.26	98.27

Table 11: Scale-free effective regularization strength $r = \beta \|\nabla_{\theta} R\|_2 / \|\nabla_{\theta} \text{CE}\|_2$ at the shared one-epoch CE-only operating point (MNIST/MLP, seed 0). The raw β grid is not a shared axis across objectives (SVIB penalizes βKL^2 , NIB is in nats, DVCCA weights reconstruction too); r is. FGIB’s row at $a=4$ vs $a=16$ shows the a^2/τ^2 coupling explicitly; NIB’s r is numerically zero across the grid (its knob does not act).

Objective	$\log_{10} r (\beta = 0.0001)$	$\log_{10} r (\beta = 0.001)$	$\log_{10} r (\beta = 0.01)$	$\log_{10} r (\beta = 0.1)$	$\log_{10} r (\beta = 1)$	1
VIB (KL, $\mathcal{N}(0, I)$)	-0.81	0.19	1.19	2.19	3.19	
SVIB (βKL^2)	2.16	3.16	4.16	5.16	6.16	
CEB (trainable prior)	-0.85	0.15	1.15	2.15	3.15	
NIB (pairwise bound)	$< 10^{-2}$	$< 10^{-2}$	$< 10^{-2}$	$< 10^{-2}$	$< 10^{-2}$	
DVCCA (KL + recon)	-0.77	0.23	1.23	2.23	3.23	
FGIB ($a=16$)	-1.25	-0.25	0.75	1.75	2.75	
FGIB ($a=4$)	-1.29	-0.29	0.71	1.71	2.71	
CentFGIB ($a=16$)	-1.25	-0.25	0.75	1.75	2.75	

epoch of every run; CEB’s prior tracks its posterior, and VIB’s posterior stays within 0.37 of zero (Figure 3(b)). To probe whether the race is an early-stopping artifact, we retrained CEB, DCEB, VIB and FGIB at $\beta \in \{1, 10\}$ for 400 epochs with early stopping disabled. The answer is nuanced and we report it as measured. CEB’s variance *does* drift, monotonically and in exactly the direction Proposition 4(i)–(ii) predicts — at $\beta = 1$ the posterior log-variance ends at -2.27 nats (-1.20 at $\beta = 10$; the prior tracks the posterior), passing -0.92 at epoch 100 — but the clamp fraction is 0.0 throughout, and -2.3 nats ($\sigma^2 \approx 0.10$) is still far from the e^{-10} clamp (Figure 4): the race is *real but slow*, and its convergence time-scale is beyond 400 epochs at these hyperparameters — the conditional status Remark 6 assigns to Proposition 4(iii). VIB shows no race at all: its fixed prior restores the posterior variance, whose log-variance holds at initialization while the accuracy collapses through the mean ($q \rightarrow \mathcal{N}(0, I)$) — predictive-information loss, not a clamp race (Remark 5). DCEB confirms the corrected sign of Corollary 1: its prior log-variance rises at $\beta = 1$ ($+0.12$ nats, posterior tracking, never down), while its $\beta = 10$ collapse occurs with the variance near initialization — the collapse mechanism is not the variance channel and remains an open, measured diagnostic. For FGIB the question is settled analytically: by Result 3 the posterior variance is pinned at τ^2 at every step — the flat zero line of Figure 4 is the fixed point itself, held for 400 epochs.

The side-head conditioning channel. Section 3.3(a) identifies the *trainable* side head A as the channel the optional variant leaves open, and predicts that the approach to a rank- K projector would show up as diverging $\kappa(A^\top A)$. Figure 3(a) plots $\log_{10} \kappa(A^\top A)$ over training, at $a = 16$, for every β : from its initialization near 3.9 the condition number rises to $10^{5.8}$ – $10^{6.8}$ at the final epoch, with transient peaks to $10^{8.3}$ — $A^\top A$ becomes *severely ill-conditioned* ($\approx 1000\times$ degradation, no monotone dependence on β). Predictive accuracy nevertheless remains stable, so the conclusion is narrow: the ill-conditioning does not hurt MNIST predictions in this experiment, but it means the pulled-back reference of Result 2 is not close to a well-conditioned frame, and the KL’s correspondence with h

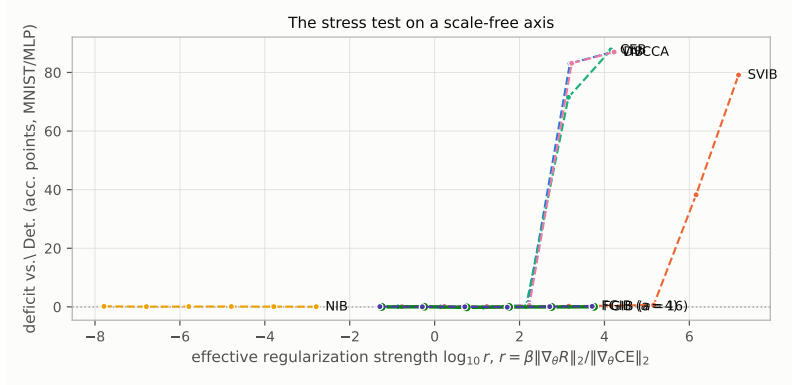


Figure 2: The stress test on a scale-free axis (MNIST/MLP). Horizontal axis: effective regularization strength $\log_{10} r$ with $r = \beta \|\nabla_{\theta} R\|_2 / \|\nabla_{\theta} CE\|_2$ at a shared one-epoch CE-only operating point (Table 11); vertical axis: the deficit in test accuracy against the deterministic backbone at that β . NIB’s r is numerically zero across the grid (its knob does not act); SVIB’s penalty βKL^2 carries different units, so its raw-grid $\beta=1$ lands at $r \approx 10^{6.2}$ — two orders of magnitude beyond any KL-nat objective’s grid — and its collapse there is a scale artifact, not evidence of fragility. Among the KL-nat objectives, VIB, CEB and DVCCA collapse between $r \approx 10^{3.15}$ and $10^{3.23}$, while FGIB reaches $r \approx 10^{3.75}$ with zero deficit.

Table 12: AgeDB under a subject-disjoint split. Identities are grouped so that no subject appears on both sides of any split boundary (60/20/20, seed 42); this removes the identity leakage of the image-level split used elsewhere in the paper, at the cost of comparability with other AgeDB literature. Validation-selected β at a fixed $a = 16$ (6-point grid), 5 seeds, CNN backbone, $R^2 \times 100$.

AgeDB (subject-disjoint)	Det.	FGIB ($a=16$)	CentFGIB ($a=16$)	FGIB-H ($A=I$, $a=16$)
$R^2 \times 100$ (val-sel. β)	21.37	35.10	35.10	33.71
selected β	—	0.001	0.001	1

is not guaranteed by anything we measured for the trainable variant. The canonical configuration needs no such caveat: $A = I$ has $\kappa = 1$ at every step and anchors h directly (Result 2). The $A=I$ and frozen- A rows of Table 13 show what the degradation costs in accuracy (little on MNIST), and Table 9 quantifies the fixed head’s price across the seven classification settings: ≈ 0.5 points per setting at the inherited anchor scale, largely recovered by re-tuning the scale — a real but bounded margin.

Fixed classifier scale. Proposition 9 requires a fixed classifier scale c , which the default `nn.Linear` does not enforce. The cosine rows of Table 13 enforce it by construction: FGIB’s sweep is unchanged (97.9–98.3 across all six β values), while VIB and CEB are not rescued (51.9% at $\beta = 1$ and 41.6% at $\beta = 0.1$ respectively), exactly as Remark 14 predicts — fixed c does not repair the trainable-prior corners; the anchor geometry, not the free classifier scale, keeps FGIB’s corner non-degenerate (Appendix D, cosine rows).

Does the geometry alone generalize? Result 3 proves the KL machinery is *never* load-bearing in this architecture, on any dataset. Table 8 asks the cross-dataset version: CentFGIB run on all 12 settings under the equal-budget protocol ($a = 16$, 6-point β grid, 5 seeds). The geometry alone generalizes: at the validation-selected configuration, CentFGIB matches FGIB($a=16$) to within 0.01 points on 10 of 12 settings and to within 0.43 on the remaining two, including all four regression settings with RFF anchors; in a nine-entry ranking the FGIB family occupies the top three slots. The empirical engine of the method is the fixed anchor geometry itself; the KL contributes the bound reading and the sweep theory (Appendix D).

Does the side head need to exist at all? The canonical configuration FGIB-H ($A=I$) removes the side head from the theory’s concerns entirely: the anchor matching acts directly on the deployed

Table 13: Confound ablations and fixed-scale checks on MNIST/MLP. Test accuracy (%) across the β grid for each variant (fixed $a = 16$ where an anchor scale applies), plus the validation-selected configuration. *DCEB* keeps the deterministic classifier path of FGIB but trains the conditional prior (CEB’s backward encoder); *TAFGIB* trains the anchor means instead of freezing them; *CentFGIB* replaces the KL with a direct MSE-to-anchor center loss; *frozen A* and $A=I$ constrain the side head; *cosine* rows use the fixed-temperature cosine classifier of Remark 14(c). Protocol identical to the main experiments (100 epochs, patience 10, 5 seeds).

Variant	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10	val-sel.
Det.	—	—	—	—	—	—	98.33
VIB	98.18	98.28	98.18	97.83	15.27	11.35	98.28
CEB	98.28	98.30	98.43	98.16	26.82	10.64	98.28
FGIB ($a=16$)	98.35	98.31	98.49	98.37	98.36	98.31	98.49
DCEB (trainable prior)	98.26	98.26	98.38	98.24	98.04	11.35	98.26
TAFGIB (trainable anchors)	98.22	98.31	98.42	98.42	98.41	98.33	98.42
CentFGIB (MSE-to-anchor)	98.35	98.31	98.49	98.37	98.36	98.31	98.49
FGIB frozen A	98.32	98.22	98.37	98.35	98.26	98.27	98.32
FGIB $A=I$	98.26	98.20	98.30	98.32	98.21	98.21	98.26
FGIB cosine	97.94	98.21	98.32	98.30	98.34	98.30	98.34
VIB cosine	97.93	98.10	98.12	96.56	51.85	11.42	98.10
CEB cosine	97.99	98.35	98.25	41.63	10.79	10.31	98.35
ETF (frozen classifier)	—	—	—	—	—	—	98.32

Method	Fixed geometry	Zero-penalty set
VIB (Alemi et al., 2017)	—	collapse (origin)
CEB (Fischer, 2020)	—	underdetermined family
center loss (Wen et al., 2016)	learned centers	no bottleneck reading
HPN (Mettes et al., 2019)	hypersphere (a priori)	—
fixed classifier (Hoffer et al., 2018; Zhu et al., 2021; Pernici et al., 2019)	simplex/ETF rows	—
FGIB (ours)	orthonormal anchors (a priori)	prescribed class codes

Table 14: FGIB against the closest fixed-geometry and prototype regularizers. “Zero-penalty set” is the set of representations where the regularizer vanishes. “Variational bound” asks whether the regularizer carries a KL/information interpretation. “Regularizer on deployed repr.” asks whether the regularized object is the representation inference consumes.

representation (Result 2). Table 9 measures the price across the complete seven classification settings at equal budget, with rank and win counts computed over the same seven complete rows. At $a = 16$, the scale inherited from the z -space sweep, the fixed head trails the trainable one by ≈ 0.5 points per setting (mean rank 3.86 of 5; FGIB 1.57, CentFGIB 1.71) — the trainable head is doing real adaptive work, and part of that work is rescaling the effective geometry. Re-tuning the anchor scale for h -space recovers most of the gap: at $a = 4$ FGIB-H takes mean rank 2.86 of 5, ahead of the backbone (3.29), with worst-case deficit against the backbone of 0.23 points and best-or-tied finishes on 2 of 7 rows (MNIST/CNN 99.29, best of the table; Cora 87.68, tied-best) — so the price of the $\kappa = 1$ closure is small and quantifiable, an accuracy-versus-certification choice whose two sides both ship in the released code. We also test the other fixed geometry the class invites: a backbone with a *frozen ETF classifier* (equiangular tight frame head, no regularizer) scores 98.32% on MNIST/MLP against the backbone’s 98.33% — a fixed classifier alone buys neither margin nor a knob, so the anchor regularization on the representation is the load-bearing ingredient (Table 13, ETF row).

Which frame? The geometry ablation. Result 2 fixes the head but not the frame: any fixed, class-separated, well-conditioned frame would close the same channel, and the fixed-prototype literature offers several (orthonormal, simplex/ETF, learned centers). Table 10 varies the frame on the *same* representation — MNIST/MLP with $A=I$, $a = 16$, the same 6-point β grid, the regularizer being either the KL or the plain MSE: orthonormal, simplex/ETF (unit norms, mutual cosine $-1/(K-1)$) and random-normalized anchors behave alike across the grid, and learned centers (TAFGIB) too — fixedness and class separation matter more than the particular frame. A frozen ETF *classifier* is not a member of this family (it regularizes nothing), which is why the ETF row of Table 13 buys no knob.

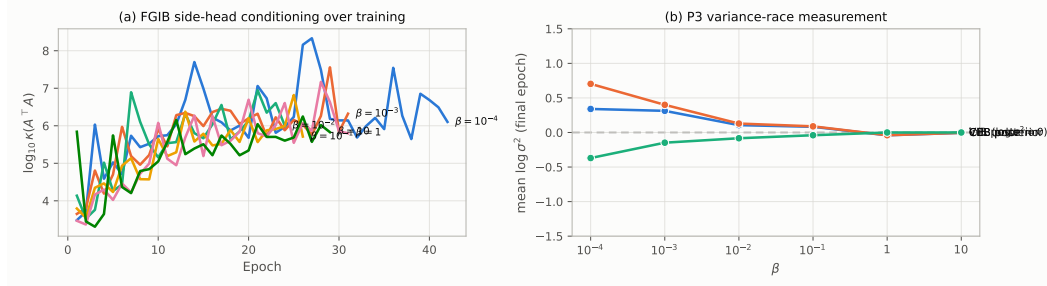


Figure 3: Diagnostics on MNIST/MLP. (a) $\log_{10} \kappa(A^T A)$ of the FGIB side head over training, one seed per β . (b) The trainable-prior variance-race measurement: mean log-variance at the final epoch against β for CEB’s posterior and prior and VIB’s posterior; the dashed line is the initialization $\log \sigma^2 = 0$.

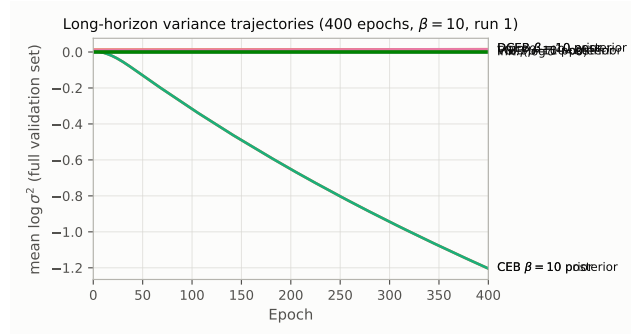


Figure 4: Long-horizon variance trajectories on MNIST/MLP (400 epochs, early stopping disabled, one seed, full validation set per epoch). CEB’s posterior and prior log-variances drift down together — the trainable-prior race in its measured direction; DCEB’s prior log-variance drifts up at $\beta=1$ — the corrected sign of Corollary 1 — while its $\beta=10$ collapse occurs with the variance near initialization; VIB’s posterior holds at the initialization — the fixed prior restores the variance, so no race exists (Remark 5); FGIB’s posterior is the flat zero line — the fixed point of Result 3, held exactly for the whole run.

AgeDB without identity leakage. The image-level AgeDB split of the main sweep leaks identity across the boundary. To check that the leak inflates magnitudes rather than creates the pattern, we reran AgeDB/CNN under a subject-disjoint split: the ordering is preserved — the backbone scores 21.37 against 35.10 for FGIB and CentFGIB and 33.71 for FGIB-H (all $R^2 \times 100$, validation-selected β) — so the leakage does not manufacture the regression claim (Table 12, Appendix D).

Anchor-scale sensitivity of the equal-budget rank. The $a=16$ equal-budget control is a post-hoc choice; Table 7 reports the fixed-anchor rank for every a in the grid: the mean rank over the 12 settings ranges from 2.00 ($a=14$, best or tied-best on 7) to 3.17 ($a=1$), with $a=16$ at 2.33 — the edge is not an artifact of the chosen anchor, and any $a \geq 8$ beats every bottleneck’s rank. The remaining tables and diagnostic figures of this section are collected in Appendix D.

F WHAT WE HAVE NOT MEASURED: FULL STATEMENT

(i) *Separating the β -effect from the early-stopping effect.* Remark 14(b) predicts that the β -dependence of the equilibrium decays with training length; our runs use patience 10, so training well past saturation and comparing converged KL across β would test it. The cosine-classifier runs of Section 4.5 enforce the fixed- c assumption of Proposition 9 but do not remove early stopping, and the 400-epoch control of that section covers the variance trajectory, not the sweep itself. (ii) *Equal budget for the other methods.* The $a = 16$ column of Table 4 gives FGIB the same 6-point budget as

the comparison; Table 7 shows the rank is stable across fixed anchor scales, not only at $a = 16$. The converse — giving VIB/CEB/DVCCA an extra temperature or variance hyperparameter dimension — is not run, so the comparison’s rank ordering is only shown to survive budget restriction for FGIB, not budget expansion for the others. (iii) *The remaining protocol asks.* All results come from one fixed 60/20/20 split per dataset (seed 42), so split-sensitivity is unmeasured — the subject-disjoint AgeDB control measures leakage sensitivity for one dataset, not split noise in general. Downstream robustness properties claimed for conditional bottlenecks elsewhere (adversarial robustness, OoD detection, calibration) are not evaluated here and carry no weight in this paper’s argument; the label-free anchor-distance score $\min_y \text{KL}(q(z|x) \parallel \mathcal{N}(av_y, \tau^2 I))$ that FGIB’s geometry makes available at inference (Remark 8) is a hypothesis, not a result.

G DATASETS AND PREPROCESSING

All splits use seed 42. Except where noted, each dataset is divided 60/20/20 into train/validation/test; classification splits are stratified, regression splits are not. Targets for regression are min-max scaled to $[0, 1]$ with a scaler fitted on the training split only, and MAE is reported in original units via the inverse transform.

MNIST (LeCun et al., 1998): 70,000 28×28 grayscale digits; the 60,000-image training partition is split into 50,000 train / 10,000 validation, the official 10,000-image test set is untouched. Inputs are normalized to $[0, 1]$.

ImageNet-100 (Deng et al., 2009): frozen ResNet-50 (He et al., 2016) layer-3 features. Two variants are used: global average pooling (1024-d, the MLP backbone) and 4×4 spatial pooling ($1024 \times 4 \times 4$, the CNN backbone). Features are split 60/20/20 (stratified) and standardized with a StandardScaler fitted on the training split.

Cora (Sen et al., 2008): the standard transductive node-classification graph (2708 nodes, 1433-d features, 7 classes) with a fixed random split (seed 42) into train/validation/test node masks; the adjacency matrix is symmetrized, self-loops are added, and the normalized Laplacian-style $\tilde{D}^{-1/2}(\tilde{A})\tilde{D}^{-1/2}$ is used. Training is full-graph (transductive), following standard GCN practice (Kipf & Welling, 2017); this is not the fixed Planetoid split, so absolute numbers are not directly comparable with published Planetoid numbers — all comparisons in this paper are internal to the same split.

IMDb (Maas et al., 2011): the official 50,000 labeled reviews (25k train + 25k test) are merged and re-divided 60/20/20 (stratified) into 30k/10k/10k. HTML tags are stripped, tokens are alphabetic words from a training-split vocabulary (approx. 72k words at length cap 250), and sequences are truncated or padded to 250 tokens. Official-split literature numbers are therefore not directly comparable; again, all comparisons are internal.

AG News (Zhang et al., 2015): frozen bert-base-uncased (Devlin et al., 2019) token features. We use the second-to-last hidden layer (layer 10) of each token, PCA-reduced to 50 dimensions (the PCA fit on training-split tokens only), sequences capped at 64 tokens; 60/20/20 stratified split over 127,600 documents.

California Housing (Pace & Barry, 1997): 20,640 samples, 8 features, 60/20/20 split (regression, unstratified); features standardized on the training split, target min-max scaled.

STS-B (Cer et al., 2017; Wang et al., 2019): the official GLUE train and dev partitions (7,249 labeled sentence pairs, similarity 0–5) are merged and re-divided 60/20/20 into 4,349/1,450/1,450 (the GLUE test partition carries no labels and is unused). Each sentence is stripped of HTML, tokenized alphabetically, and truncated to 124 tokens; the pair is joined by a [SEP] token into a single sequence of length ≤ 250 . The vocabulary is built from the training split (approx. 10k words). Embeddings are GloVe-6B-100d (Pennington et al., 2014), initialized and frozen; the LSTM has a single 256-d hidden layer and max-pooling readout.

ZINC-12k (Dwivedi et al., 2023): the official benchmarking-gnns split (10,000 train / 1,000 validation / 1,000 test). Node features are 28-d one-hot atom types (bond types are unused, matching the benchmarking GCN baseline); the adjacency is symmetrized, self-loops added, and degree-normalized as for Cora; batching uses block-diagonal adjacency (batch size 128). The target is the penalized logP, min-max scaled on the training split; MAE is reported in original units (our GCN baseline reproduces the literature value to within the expected margin).

AgeDB (Moschoglou et al., 2017): the official full release, 16,488 images of 567 subjects aged 0–96, resized to RGB 64×64 and normalized with ImageNet statistics; target age min-max scaled on the training split. **Split caveat:** the default split is image-level 60/20/20, under which 562 of 567 subjects have images on both sides of the train/test boundary — identity-specific age cues can leak, and the AgeDB numbers should be read with that in mind. A subject-disjoint split (grouping all images of a subject into one partition, then splitting subjects 60/20/20 with seed 42) is implemented in the released code and *is run* as a control in this paper: under it the objective ordering is preserved (Table 12), so the leak inflates AgeDB’s magnitudes but does not create the pattern.